Innovative learn-by-doing digital education and attitudes towards STEM: an RCT in Italian high schools*

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Abstract

We conducted a cluster-randomized controlled trial to test the effectiveness of creative STEM activities on the propensity to enroll in a STEM university course and career aspirations in a sample of 573 high school students in Italy. Classes (42 classes across 5 schools) were randomly assigned to treatment or control; students in treated classes were offered the possibility to participate in activities run by FabLabs, non-profit digital fabrication laboratories with an innovative pedagogical approach grounded in learn-by-doing, problem-solving skills, and creativity. The activities covered courses on 3D printing, laser cutting, and programming. Participation was voluntary but counted for credit; we estimate Intention-to-Treat effects. We find that access to these courses increased students' intentions to pursue STEM majors at university (6 p.p., +17%) and STEM careers (7.4 p.p., +40%), possibly through increased STEM self-efficacy (7.3 p.p., +22%), namely students' beliefs in their ability to pursue STEM studies.

Keywords: STEM, university choice, digitalization

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1 Introduction

Recent technological progress and automation are profoundly transforming the demand for labour. While highly educated workers are generally advantaged, skills that are complementary to digital technologies – in the sense that they allow workers to use, manage, and build upon new technologies rather than being replaced by them – are increasingly valued in the labour market. While these skills are typically advanced and STEM-oriented, analytical, and non-repetitive, they also involve complex problem-solving, creative thinking, and collaboration skills, implying a high degree of adaptability to rapidly changing labour markets (Black et al., 2021; Lassébie and Quintini, 2022; Acemoglu and Restrepo, 2019, 2018, 2020; Dell'Acqua et al., 2025).

From a policy perspective, this creates the double challenge of increasing the number of STEM graduates while also equipping them with soft skills that enhance long-term adaptability. At the same time, however, the supply of STEM graduates loses students at several steps of the education process (a "Leaky pipeline", Wickware, 1997). Secondary education curricula worldwide struggle to include the use of modern digital technologies to a sufficient degree (OECD, 2020), potentially decreasing the attractiveness of STEM majors among high school students. Later on, not all STEM graduates transition into STEM occupations, especially in the case of women (Meoli et al., 2024). Indeed, while the undersupply of STEM professionals concerns both sexes, a large literature has shown that attracting women into STEM is particularly challenging, further contributing to the gender gap in wages and employability (e.g., Jiang, 2021; Charlesworth and Banaji, 2019). An important and understudied policy question, therefore, is whether early exposure to creative, applied uses of digital technologies can shift students' educational and occupational intentions before post-secondary choices are made.

In this paper, we test whether creative STEM activities offered to high school students can increase their interest in pursuing STEM university studies and a career in STEM-related fields. Within a cluster-randomized controlled trial in Italian high schools conducted in the 2021-2022 academic year, we randomly assigned classes to either treatment or control, offering treated classes the possibility to participate for credit in activities run by FabLabs, organizations that adopt a learning-by-doing pedagogical approach in teaching the use of digital fabrication tools, and whose activities are grounded in fostering creativity, collaboration, and problem-solving. Originally established by the Massachusetts Institute of Technology (MIT) in the early 2000s, FabLabs are now diffused worldwide, both as independent laboratories and within schools, universities, and corporate R&D departments (Oppedisano, 2024).

In our trial, students are trained to use FabLab equipment, including 3D printers, laser cutters, and CNC milling machines, and software, including 3D modeling and coding, to bring their ideas to life by creatively solving actual problems rather than learning mnemonically. Students used these tools to design and build robots equipped with sensors to interact with the environment, low-cost microscopes for their school, new benches for a local park, and replicas of local sculptures.

FabLab activities are theoretically well suited to affect STEM choices because they target several determinants emphasized in the literature: perceived self-efficacy, intrinsic interest in STEM tasks, information about what STEM work can look like, and the perceived social usefulness of STEM-related activities. We formalize these channels in a conceptual framework, developed in Section 3, that distinguishes between beliefs – especially students' perceived ability to engage with STEM – and preferences – especially students' intrinsic interest in STEM activities and careers.[1]

The experiment was conducted in Italy, a setting where the policy relevance of this question is particularly high. Italy combines historically low innovative capacity (Colombo and Delmastro, 2002) with labour-market pressures similar to those observed in other industrialized economies, where digital-intensive employment is growing especially in non-repetitive, creative, and cognitive tasks (Cirillo et al., 2021). Yet Italy remains among the European countries with the lowest tertiary enrollment rates, and the shortage of STEM graduates concerns both men and women.[2]

Our sample consists of 710 students from 5 high schools in Italy, of which 573 have complete baseline information and constitute the analysis sample. We randomize access to FabLab activities at the class level (42 classes), so that treated and control students do not share the same classroom, preventing contamination between groups. At the beginning of the 2021-2022 school

[1] This distinction is also consistent with the broader literature on women's under-representation in science. Reviewing over 470 studies, Avolio et al. (2020) emphasize individual factors, such as perceived self-efficacy and attitudes toward science (see also Scott and Mallinckrodt, 2005); labour-market factors, such as information about possible career types; educational factors, including academic performance, perceived skill requirements, and pedagogical approaches that may reinforce negative attitudes toward STEM (see also Blickenstaff, 2005); and social factors, especially the perceived possibility of being socially useful as a STEM professional (see also Sax et al., 2016).

[2] Italy has historically been characterized by a relatively low tendency to innovate (Colombo and Delmastro, 2002). However, Routine-Biased Technological Change is highly relevant in the Italian labour market: digital-intensive employment among highly skilled workers is growing especially in non-repetitive, creative, and cognitive tasks that are less exposed to automation (Cirillo et al., 2021). In this context, digital entrepreneurship based on creative skills and intellectual property may be more resilient to negative economic shocks (Khlystova and Kalyuzhnova, 2023), while young innovative firms are more likely to thrive when founders have scientific, technological, or industry-specific expertise (Colombo and Grilli, 2005). These patterns make the limited supply of STEM skills especially salient: Italy remains among the EU countries with the lowest tertiary enrollment rates, especially among men (Eurostat, 2025a), and the share of male STEM graduates in the population remains substantially below European figures despite men's higher propensity to choose STEM conditional on enrolling in tertiary education (e.g., Xie et al., 2015; Jiang, 2021; Ceci et al., 2014; Eurostat, 2025b).

year, students in the treated group were offered to join the FabLab STEM activities, whereas those in the control group could only choose alternatives unrelated to STEM. In both cases, the activities count as "PCTO" credits ("*Percorsi per le competenze trasversali e l'orientamento*"), required by law for all high school students and linked to internships, extra-curricular courses, or social work meant to orient students in the labor market.

We conduct our analysis at the individual level to leverage detailed student-level baseline information.[3] Before randomization, we administered a baseline survey covering pre-treatment attitudes towards university, STEM majors, and career aspirations, as well as parental educational attainment, field of specialization, and occupation. We integrate these data with school administrative records on students' GPAs. At the end of the school year, we conducted a second survey to measure outcomes. For treated students, we also track actual enrollment in FabLab courses using administrative records from both schools and FabLabs.

We estimate intention-to-treat effects on students' university intentions, STEM-study intentions, STEM-career aspirations, and the potential intermediate outcomes predicted by the conceptual framework, including self-efficacy. We then complement the main experimental estimates with causal forests, which we use to further explore heterogeneity in predicted treatment effects at the individual level and to run several robustness checks.

We find that while offering FabLab activities does not impact the overall intention to attend university, it increases the intention to pursue a STEM profession by 7.4 percentage points, a 40% increase relative to baseline. Accounting for the overall uncertainty of attending university at all, intentions to pursue STEM-related majors at university increase by 6 percentage points, equivalent to a 17% increase. The effect on STEM-university intentions is accompanied by an increase in STEM self-efficacy of 7.3 percentage points, or 22% relative to baseline. The individual-level heterogeneity analysis performed via causal forests is consistent with this channel: students with larger predicted gains in self-efficacy also display larger predicted gains in STEM-university intentions. By contrast, larger predicted gains in self-efficacy are not associated with larger predicted gains in STEM-occupation intentions, suggesting that career aspirations may respond through a distinct pathway, more compatible with direct exposure to STEM as a creative, applied domain that broadens students' conception of what STEM-related occupations can look like.

Consistent with previous interventions studied in the literature, male students drive the observed effects. For females, we detect a smaller and less precise effect on intentions to pursue

[3] Since 137 students were absent from school on the date of the baseline survey and thus had missing baseline information, they were excluded from the final analyses.

STEAM studies, which comprise STEM plus digital-intensive artistic fields such as interior design, suggesting that the creative component of FabLabs may generate interest among some female students. This pattern is policy-relevant because the intervention was designed with voluntary compliance, mirroring the actual organization of PCTO credits in Italian schools. Our analysis of compliance, which was directly caused by the availability of alternative courses across schools, suggests that compulsory attendance of creative STEM activities may be more beneficial for females and for students with lower initial self-efficacy.

The paper contributes to three related literatures. First, it adds to research on STEM educational choices by providing experimental evidence on an intervention that acts before university enrollment and targets both educational and occupational intentions. A vast body of observational literature has focused on the determinants of choosing a STEM major, especially among women. In their review of US-based studies, Xie et al. (2015) find that while socioeconomic background matters for attending university in general, social-psychological factors are relatively more important in defining whether students pursue a STEM major. Our findings are consistent with this distinction: FabLab activities do not affect general university intentions, but shift STEM-specific intentions and self-efficacy.

Second, the paper contributes to the literature on STEM interventions in secondary education. Existing causal evidence has focused mainly on increased curricular exposure to mathematics and sciences (Black et al., 2021; De Philippis, 2023) or exposure to role models (e.g., Breda et al., 2023; Porter and Serra, 2020). These interventions can operate through academic performance, information, and self-efficacy. FabLabs instead change the pedagogical mode of exposure to STEM by emphasizing making, experimentation, collaboration, and creative problem-solving.

Third, the paper connects to a growing literature on inventive activities, technological imprinting, and entrepreneurship, mostly focused on adults. Dalziel and Basir (2024) show that brief inventive activities based on practical learning can shape technological trajectories among *de novo* entrepreneurs, while evidence from Italy links scientific and technological education to innovative entrepreneurship (Colombo and Grilli, 2005; Colombo and Piva, 2020). Del Carpio and Guadalupe (2022) randomized an informational campaign about upcoming coding courses for low-income women in Peru and Mexico, including access to a female role model. We extend this line of work to secondary schools, asking whether creative STEM exposure can shift intentions before students choose tertiary education and occupational paths.

To our knowledge, this is the first randomized evaluation of learn-by-doing creative STEM activities in secondary schools on attitudes towards university, STEM studies, and STEM careers.

The paper proceeds as follows: Section 2 describes the institutional context, Section 3 develops the conceptual framework linking FabLab activities to beliefs and preferences, Section 4 describes the sample composition, Section 5 turns to the experimental design, the treatment, and the outcomes, Section 6 details the empirical strategy, Section 7 presents the main results and explores heterogeneity, compliance, and potential mechanisms. Finally, Section 8 concludes the paper.

2 Institutional context

2.1 The Italian school system and university enrollment statistics

Our paper examines the impact of the FabLab pedagogical approach in a sample of upper-secondary schools in Italy. The Italian education system is structured into three main levels: primary education, lasting five years (ages 6 to 11); lower secondary education, lasting three years (ages 11 to 14); and upper secondary education, which covers ages 14 to 19.[4] While primary and lower secondary schools are characterized by a rather strong standardization of curricula, upper secondary education is marked by higher curricular differentiation (Ballarino and Panichella, 2021).

At the age of 14, students are required to choose among several educational tracks. The most prestigious among these are the academic-oriented schools (*licei*), which typically enroll the highest-achieving students and provide a better preparation for university studies (Triventi et al., 2021). In contrast, vocational schools (*istituti professionali*) aim at training students for manual occupations in the service and industrial sectors. Technical institutes (*istituti tecnici*) offer specialized training in fields such as mechanics, electronics, and business management, aiming to facilitate both direct entry into the labor market and access to higher education, even if they are considered less academically demanding than *licei*.

Despite these differences, all tracks confer eligibility for university admission. Nevertheless, different tracks place varying emphasis on specific curricular subjects, potentially affecting the ability to pursue university studies in specific fields. Only a minority of subjects (e.g., mathematics, Italian, English) are present in each curriculum, even if with a varying degree of intensity. Within each educational track, there are internal subdivisions: the *licei* are primarily distinguished by the relative weight given to STEM disciplines versus the arts and humanities (e.g., *licei scientifici* have a higher focus on STEM disciplines in contrast to *licei classici* focused on traditional humanities; *licei linguistici* focus on foreign languages and *licei artistici* on arts); the

[4] 3-years vocational courses are chosen by a small minority of students and will not be discussed here.

focus of *istituti tecnici* can be either on business-related or on technical fields; while the *istituti professionali* vary between care-related and technical subjects (Contini et al., 2025). Tertiary education is more homogeneous: unlike in countries with dual academic–vocational systems, Italy delivers almost all tertiary education through universities, with limited short post-secondary vocational options (Ulicna et al., 2016).

Tracks and sub-tracks in high schools exhibit a notable segregation on the basis of students' sex and social origins. Empirical evidence highlights considerable social homogeneity within schools and classrooms (Barone et al., 2021): *licei* predominantly attract students from highly educated families and the upper classes; *istituti professionali* mainly serve students from less educated and frequently migrant backgrounds; while *istituti tecnici* occupy an intermediate position on the socio-economic gradient, as they are conceived as a viable alternative to students who are not sure whether to continue to university after the diploma. Similarly to global trends, tracks that place greater emphasis on STEM or technical subjects tend to enroll a higher proportion of male students (Contini et al., 2025).

As a result, a system that formally grants access to university to all students with a five-year upper-secondary school diploma generates distortions in university enrollments. In Italy, actual university enrollment rates remain relatively low compared to other European countries—both in the overall working-age population and among younger generations, with only modest improvements over time, representing an ongoing policy challenge (Eurostat, 2025a). This outcome is influenced mainly by the fact that only students from *licei* pathways tend to perceive themselves as sufficiently prepared to undertake university, resulting in overall low self-efficacy. Specific to the Italian context, instead, male enrollment rates are especially disadvantaged, displaying a wider educational gender gap compared to the European average. A similar gap in comparison with the rest of Europe is also evident for the field of study: the share of STEM graduates in the young working population (20-29) is in line with the EU averages for females (approximately 15%), but much lower for males – 22% in Italy *vs* 29% in the EU (Eurostat, 2025b). More detailed statistics on graduates by country, field, and gender are reported in Section B of the Appendix.

Summarizing, Italy stands out in Europe for having an overall low university enrollment rate; the share of STEM graduates is especially low among males although, in absolute numbers, female STEM graduates are less than males, similar to other countries. In other words, the lack of STEM graduates in Italy represents an ongoing and particularly relevant policy challenge, since STEM studies are rarely chosen also by males, who display higher STEM enrollment rates in other countries.

2.2 FabLabs

FabLabs (namely, fabrication laboratories) are small-scale workshops that offer access to tools for digital fabrication and adopt a hands-on pedagogical approach. This concept was created at the Massachusetts Institute of Technology (MIT) in 2002 when digital fabrication tools, including 3D printers, CNC machines, and electronics, were packaged in a standardized low-cost lab (Blikstein and Krannich, 2013). The founder of the FabLab concept, Prof. Neil Gershenfeld, describes this digitalization of fabrication as the process "where you do not just digitize design, but the materials and the process" (Solon, 2013), so digital fabrication has the mission of "bringing programmability to the real world". FabLabs projects, which have now spread in educational and entrepreneurial contexts all over the world, range from jewellery to furniture and all the way up to entire houses (Ferracane, 2020).

FabLabs began projects in primary and secondary education in 2008 thanks to Stanford University's FabLab@School project, with a specific focus on incorporating the "making" into school curricula. The pedagogical approach is based on "playful experimentation", i.e., the practical use of tools and materials to make the learning experience more engaging (Regalla, 2016). The specific activities offered by FabLabs in our study are described in Section 5.1.

3 Conceptual framework: beliefs, preferences, and educational choices in STEM

Student enrolment decisions involve three conceptually distinct drivers: subjective expectations about the costs, returns, and probability of success associated with each option; intrinsic preferences for the activity or subject itself; and self-efficacy, the student's belief in her own ability to succeed in a given domain. We briefly review three complementary frameworks that, taken together, let us separate these drivers and identify the channels through which a creative STEM intervention such as FabLab courses can plausibly operate.

Rational-action frameworks conceptualize education as an investment, either in human capital (Becker, 1964) or in the avoidance of social demotion (Breen and Goldthorpe, 1997), and emphasise three parameters for decision-making: the costs, the expected returns, and the probability of success of each option.[5] Subjective expectations on costs, returns, and the probability of success are predictive of whether students continue to higher education (Breen and Yaish, 2006).

[5] These frameworks have long been formulated under bounded rationality (Simon, 1955), with decision-makers relying on heuristics and limited information-seeking rather than on fully Bayesian processing (Tversky and Kahneman, 1974; Kahneman, 2011; Morgan, 2005).

They fall short, however, in explaining the horizontal differentiation of educational choice—which field or major students choose at a given level—even when elicited directly (Gabay-Egozi et al., 2010; Arcidiacono et al., 2012). A second parameter is therefore needed alongside expectations. The expectancy-value model by Eccles and Wigfield (2002) introduces it as intrinsic value: students engage with a domain when they both expect to succeed in it and value the activity itself. Crucially, these two components vary independently—a student may find science interesting yet doubt her ability to succeed in it, or feel competent in mathematics without being drawn to it. Social Cognitive Career Theory (Lent et al., 1994) extends this framework to career development, placing self-efficacy at the centre of a process whereby efficacy beliefs shape outcome expectations and interests, which in turn guide career choice actions.

Across these three frameworks, subjective beliefs about one's self-efficacy (Bandura, 1977, 1997) play a particularly prominent role. Math self-efficacy is among the strongest predictors of entering a STEM major (Wang, 2013), and self-efficacy gaps help explain the persistent gender imbalance across STEM fields (Cheryan et al., 2017). These beliefs, however, are often inaccurate: students systematically misjudge their own ability (Usher, 2005) and report substantial uncertainty when asked to do so (Abbiati and Barone, 2017). The "adolescent econometricians" (Manski, 1993) thus make consequential decisions not only under imperfect information about costs and returns, but also about their own potential. This uncertainty is not a destiny: students update their beliefs, and revise preferences and prior decisions, when new information becomes available (Wiswall and Zafar, 2015; Barone et al., 2017).

The creative pedagogical approach undertaken by FabLabs, as detailed in Section 5.1, can plausibly operate through both channels. On the belief side, FabLab activities have students use digital fabrication tools (3D printers, laser cutters, and CNC machines, among others) to design and build functional objects, providing hands-on mastery experiences that can directly shift self-efficacy. On the preference side, FabLab activities expose students to a qualitatively different type of STEM than what they encounter in school curricula. Most students have never experienced digital fabrication, and the creative, collaborative, and applied nature of these activities—in which STEM skills serve as tools for solving real problems and producing tangible objects—may broaden students' conception of what STEM encompasses and increase their interest in it.

This dual-channel mechanism distinguishes FabLab activities from previously studied interventions in this area. Increased curricular exposure to traditional STEM subjects (Black et al., 2021; De Philippis, 2023) provides more hours of the same type of instruction students already receive: the content is familiar, and the mechanism runs primarily through improved academic performance and deeper familiarity with existing subjects. Role model interventions (Breda et al., 2023; Porter and Serra, 2020) provide career information and vicarious learning, showing students that "someone like me" can succeed in STEM. Informational interventions provide students with information about choice parameters or reveal their stereotypes (Ballarino et al., 2022; Owen, 2023). FabLab activities that adopt a creative pedagogical approach are qualitatively different: they introduce students to an entirely new domain of STEM that can foster interest in this subject, while simultaneously providing mastery experiences through hands-on activities, which can enhance self-efficacy. This setup is consistent with recent evidence about the potential of practical activity for the reinforcement of the effectiveness of belief-based interventions. For instance, Yeager and Walton (2011) show that brief interventions targeting psychological mechanisms such as self-efficacy can produce lasting effects when they feed into recursive processes – for instance, if students who gain confidence in STEM subsequently seek out further STEM experiences, reinforcing the initial shift.

This framework, therefore, generates three testable predictions. First, we should observe an increase in STEM self-efficacy, students' beliefs about their ability to pursue STEM education, driven by the mastery experiences that FabLab activities provide. Second, intrinsic interest in STEM may also increase, as exposure to a novel and engaging STEM experience can broaden students' appreciation for the field. Third, if beliefs and preferences shift, downstream educational intentions (such as the willingness to pursue a STEM major or a STEM career) may follow, because students who believe they can succeed in STEM and find it appealing are more likely to consider it as a viable option.

4 Sample composition and summary statistics

Students who participated in the program were enrolled in five different Italian high schools, from grade 10 (the second year in the upper-secondary school cycle) to grade 13 (the fifth and last year).[6] The five schools – each one associated with a different FabLab – are located in different cities in the North, Center, and South of Italy: Schio, Verona, Mantua, Ancona, and Agrigento. A map illustrating the geographic location of each city is presented in Figure A.1, in the Appendix. Schools were included because they voluntarily responded to invitations from the partner FabLabs, which identified potential participants through their local knowledge of

[6] Specifically, 10th graders accounted for 58.1% of the analysis sample, 11th graders 7.1%, 12th graders 17.1%, and 13th graders for 17.6%. Considering instead the initial sample – i.e., including students who were absent on the day of the baseline survey – 10th graders account for 57.7% of the analysis sample, 11th graders 8.87%, 12th graders 17.0%, and 13th graders 16.3%.

schools in each city or town, independently of school track.[7] Three tracks are present in the schools of our sample: technical schools (*istituto tecnico*), arts schools (*liceo artistico*), and scientific schools *(liceo scientifico)*, the latter specialized in mathematics and sciences. In all cases, FabLab courses complement the curricula. For the first two in particular, although their curricula place greater emphasis on practical subjects than other schools, they do not include the specific skills taught in FabLab activities (e.g., 3D printing, laser cutting, programming). On the other hand, these schools exhibit enrollment rates in university below the average (especially in STEM subjects), and attract the students who middle school teachers and families deem to be the least likely to follow university studies, as discussed in Section 2.1.

Table 1 presents summary statistics for the 573 students for whom we observe baseline characteristics and conduct our analyses. Our sample comprises slightly more males than females. Most students have both parents employed, but with generally low educational attainment (only one third of students have at least one parent with tertiary education). Approximately 41% of students would choose a STEM faculty, while only 25% plan to pursue a STEM professional career.

5 Experimental design

This section describes in detail the stratified experimental design, our treatments and outcomes, following our pre-registered analysis plan, which is available at the AEA RCT registry, trial number 8407 (Dominici et al., 2025). Any deviations from the pre-analysis plan are noted in the text and in Section C of the Appendix.

5.1 Treatment

Fablab activities covered in our study were conducted between September 2021 and May 2022. They were carried out in each school by instructors employed by the local FabLab since, as detailed in Section 5.3, randomization is carried out within each school, and the analysis is stratified by school.[8] However, all FabLab instructors received the same formation from the

[7] Voluntary school participation imposes a scope condition on external validity, but does not threaten the internal validity of the causal estimates: random assignment of classes to treatment or control occurs within each participating school (*stratum*), independently of any school-level characteristic, including their technical or academic track. Our results should therefore be interpreted in the context of schools that voluntarily agree to host hands-on STEM activities run by external partners, a scope condition shared by most school-based field experiments in the economics of education (for a recent Italian example of an external-partner extra-curricular STEM activity in voluntarily participating schools, see Carlana and Fort, 2022).

[8] We also include school fixed effects in our main analysis, which will also capture the variation brought by different FabLab educators.

Table 1: Summary statistics pre-treatment

	Mean	SD	Min	Max	N
Female (=1)	0.38	0.48	0.00	1.00	573
No Siblings (=1)	0.17	0.38	0.00	1.00	573
Age	16.53	1.31	14.00	22.00	573
First Born (=1)	0.51	0.50	0.00	1.00	573
Parent Working (=1)	1.77	0.43	0.00	2.00	573
Parent University (=1)	0.34	0.47	0.00	1.00	573
University (=1)	0.44	0.50	0.00	1.00	573
STEM University (=1)	0.41	0.49	0.00	1.00	573
STEM Ability (=1)	0.34	0.47	0.00	1.00	573
STEM Interest (=1)	0.41	0.49	0.00	1.00	573
GPA	7.21	0.98	0.00	10.00	573
STEM Career (=1)	0.25	0.44	0.00	1.00	573
Fav. Subject STEM (=1)	0.52	0.50	0.00	1.00	573
Least Fav. Subject STEM (=1)	0.41	0.49	0.00	1.00	573

Notes: The table displays summary statistics for our analysis sample, comprising 573 students for whom we observe baseline survey answers. STEM Ability and STEM Interest are self-reported measures of perceived ability and interest in STEM at university, respectively. STEM Career denotes self-reported interest in a STEM professional career. GPA is self-reported.

investigators and applied the same pedagogical approach. Moreover, the contents of the activities were designed and managed centrally by Fondazione Edulife,[9] meaning they are harmonized across the participating FabLabs.

All the FabLab activities offered as part of our study lasted approximately 25 hours[10] and share a learning-by-doing approach in teaching the use of digital fabrication tools (e.g., 2D and 3D modelling, 3D printing, laser cutting), group work, an informal and relaxed learning environment, playful activities, and a learning model based on peer-to-peer interactions and aimed after a specific, self-contained project. In other words, all the involved students learned to use digital fabrication tools creatively to solve actual problems or build something of practical use, often involving the wider local community, rather than studying mnemonically (Resnick, 2017; Ferracane, 2020) – a pedagogical model rarely adopted in Italian high schools.

While these features were common to all the FabLab courses, the specific project varied across FabLab-school pairs, corresponding to strata in our experimental design. In Verona, students built low-cost microscopes for their school using common objects (wood cuts and webcams),

[9] As part of the Fabschool project funded by Fondazione Cariverona.

[10] Light-touch interventions can produce lasting effects on educational trajectories when they operate through psychological mechanisms, such as self-efficacy and sense of belonging, that feed into recursive processes in educational settings (Yeager and Walton, 2011).

resorting to digital fabrication for their assembly (Figure 1**B.**). In Schio, digital fabrication was aimed at building robots that could interact with the surroundings through the creation and application of sensors. In Mantua, students learned photogrammetry (a technique used to extract measurements and 3D information from photographs, to be used as input for 3D modelling and printing) to create replicas of statues from the local monumental cemetery (Figure 1**C.**). In Ancona, students learned 2D modelling and used CNC milling machines and laser cutting to create benches that were eventually placed in the local park, structuring the design phase with consideration of both the practical construction of the benches and their decorations, and the needs of final users (Figure 1**A.**). Finally, in Agrigento, digital modeling techniques were used to design the decorations of T-shirts, which were eventually realized with laser cutters and digital embroidery machines.

Alternative activities for control-class students. Classes assigned to control were offered alternative activities chosen by their school to complete PCTO credits, mirroring the real-world setting that a policymaker would face by implementing FabLab courses as a PCTO activity. Across the schools in our sample, three alternatives were proposed: no alternative activity offered in the same academic year (*"nothing"*, 7 classes/92 students); an internship in a non-STEM company (*"intern"*, 4 classes / 40 students); and courses covering safety on the job and basic financial literacy delivered in school (*"safety/finance"*, 4 classes / 32 students). Importantly, PCTO credits can be accumulated across the last three years of the upper-secondary education cycle rather than on a yearly basis. As a consequence, students in treated classes who chose not to enrol in FabLab activities were not obliged to attend any alternative in the same academic year, as they could simply postpone PCTO accumulation to a later year, and the same holds in principle for control-class students whose school offered no alternative.[11] The type of alternative is highly correlated with both school and grade: each of the five participating schools offered exactly one type of alternative, and the alternative also varies systematically with grade (grade-2 classes always faced "nothing", grade-3 classes always faced "intern", grades 4–5 split between "intern" and "safety/finance").

5.2 Outcomes

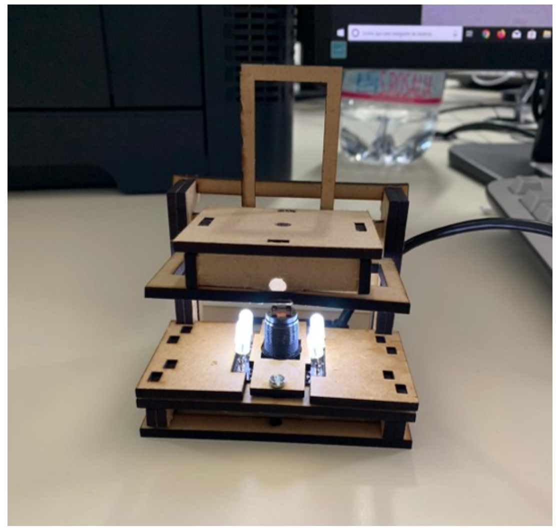
We focus on three different families of primary outcomes: attitudes towards STEM, university choice, and career aspirations.[12] Being of a subjective nature, all outcomes were self-reported

[11] Among classes assigned to treatment, some were in second grade: nevertheless, in their specific case, FabLab courses counted towards PCTO credits accumulation.

[12] In our Pre-Analysis Plan, we did not distinguish between attitudes towards STEM and University choices, and grouped them together under the label of "Interest in STEM subjects". Instead, we included preference over

Figure 1: Examples of FabLab courses' outputs

(A.) Ancona: benches



(B.) Verona: microscopes

(C.) Mantua: cemeterial statues

and measured both at baseline and endline via surveys carried out individually by each student, in class, under the supervision of investigators and instructors. The full text of both the baseline and endline questionnaires is reported in Section E in the Appendix.

Attitudes towards STEM We measured attitudes towards STEM subjects with two questions. The first was aimed at detecting the interest of the students in this possibility: "At university, would you be interested in taking STEM courses?", followed by a reminder of what fields of study are classified as STEM. The second, instead, was aimed at detecting self-efficacy in case of an enrollment in a STEM field: "If you went to university, would you feel able to follow/take a course in STEM subjects?" Both of these questions could be answered in a binary form (yes/no).

University Choice Intentions We test whether human capital decisions are altered by FabLab activities. We expect students may shift their intentions towards pursuing university education, and, more specifically, STEM courses at university. The intentions to attend university, and in which field, could be altered at both the intensive and extensive margins. Namely, it could affect both students who were undecided regarding studying at university and steered towards enrollment following our intervention, or students who wanted to enrol in non-STEM different majors who updated their preferences following FabLab courses. Taking this distinction into account, we collected two measures of university choice: we first asked students whether they wished to enrol in university, and then asked them to indicate all the potential fields of study. We coded as STEM fields the following options in our survey, corresponding to ISCED-13 codes 05, 06, 07, the standard STEM study field classification (UNESCO Institute for Statistics, 2015): engineering, information technology, biology, physics, astronomy, mathematics, architecture, earth sciences, computer sciences, and chemistry. Finally, we created a binary variable equal to one if a STEM field was selected. Given the emphasis on artistic creation and design in the proposed FabLab activities, as a secondary exploratory outcome, we also repeat the analysis considering STEAM rather than STEM, namely including design-oriented studies (e.g., industrial and interior design), which now routinely employ digital design tools.

Career Aspiration Finally, we assessed the intention to pursue STEM careers through open-ended questions. We first asked students if they had ideas about their future job and, subse-

school subjects as a pre-registered outcome, meaning only Mathematics. Given the nature of FabLab courses and pedagogical approach, and later insights, we still maintained three families of outcomes, but we: (i) Omitted preference for mathematics, which is, however, available upon request; (ii) Distinguished between attitudes towards STEM, meaning interest and perceived self-efficacy, which can be seen as intermediate outcomes, and actual intentions to pursue university studies, and if so, whether in a STEM field.

quently "If yes, what job do you imagine", in the form of an open question.[13] We then asked more specifically: "Imagine yourself as an adult. Realistically, what job do you think you will do?". We used the latter question to build our outcome variable on career aspirations, since it was aimed at eliciting realistic aspirations.[14] In our main classification, we classified jobs into two categories: STEM professionals versus other professions, according to the taxonomy developed by the 2014 Occupational Outlook Quarterly, U.S. Bureau of Labor Statistics (Vilorio, 2014). STEM professionals are individuals aiming at a STEM-related job that requires a tertiary degree in a STEM field. Following best practice, each open answer that required classification has been independently coded by two of the authors of this paper. In the first round of coding, we reached a level of agreement higher than 95%. The few disagreements were resolved by a third coder, chosen from the other authors of this paper.[15] Given the peculiarity of the FabLab experience and the presence of students from artistic school tracks, also for this outcome we built a secondary indicator that expands the definition of STEM jobs by also adding artistic jobs that require a technical component, thereby creating the STEAM category.[16]

5.3 Randomization, balance and attrition

The final sample used in our analyses consists of 573 students in 5 schools, comprising 42 classes. They were surveyed twice: once at the beginning of the school year and once at the endline, after FabLab courses had taken place. The reason we only use part of the full sample of 710 students is that we observe baseline characteristics necessary for our model estimation only for these 573. While every student observed at baseline is also observed at endline (i.e., we had no attrition in the common sense), there are 137 students out of the initial 710 for whom we only observe endline answers: this "inverse" attrition is due to absence from school on the day of the baseline survey, or to a missing or wrong name in either survey, which prevents the match. We could test for systematic differences between our analysis sample ($N = 573$) and these excluded subjects ($N = 137$) only in those time-invariant characteristics that are not affected by treatment, which we can extract from the endline survey: sex, parental education

[13] This first question was also asked in the baseline questionnaire and used as a control variable in our regressions, to increase estimates' precision and power.

[14] The results are very similar when building the outcome using the former question (results available by request to the authors).

[15] The full taxonomy is freely accessible at https://www.bls.gov/careeroutlook/2014/spring/art01.pdf.In case of ambiguous answers (e.g., "*electronic*"), we took advantage of the field of study and of the likelihood of enrolling in university in the coding procedure. In case of multiple jobs listed (e.g., "*Engineer, surgeon*"), we coded the job as STEM if at least one of the occupations listed could be classified as such. We included among STEM jobs "Commander of the Navy", since this job position in Italy requires a University-level degree in marine engineering.

[16] In the STEAM category we included the jobs "interior designer" and "photographer".

and occupational status, whether the student has siblings and whether he/she is the firstborn. We found two significant differences, although small in magnitude: students excluded from the analysis were more likely to be male (difference in means equal to 0.18, $p = 0.00$) and to have no siblings (difference in means equal to 0.09, $p = 0.01$). Given males' higher interest in STEM, this potentially imposes a downward bias on our estimates, making them conservative.[17] Most importantly, and in line with the notion that missed matches were due to random school absences, this "inverse" attrition is not correlated with treatment status.[18]

Our experiment is a cluster-randomized controlled trial and the unit of randomization is the class, within schools (strata). The class-level probability of assignment to treatment is equal to 0.7, which results in 28 treated classes (409 students) and 14 control classes (164 students). We test balance in baseline covariates and show the results in Table 2. The tests are performed by estimating the following model for every covariate X_k with $k = 1, \ldots, K$:

$$X_{k,i} = \alpha_{0,k} + \alpha_{1,k} Treat_i + \alpha_{2,k} School_i + \eta_{k,i}, \qquad i = 1, \ldots, N,$$

where $Treat_i$ takes value 1 if student i belongs to a treated class, and 0 otherwise.

As displayed in Table 2, randomization manages to equalize baseline characteristics except for one covariate: treated students are less likely to report STEM subjects as their favourite. Similarly to potential spillovers between treated and control classes within the same school this imbalance could, if anything, make our estimates more conservative by introducing downward bias. In any case, we adjust for all the covariates measured at baseline to control for this difference.

[17] We conducted the test by testing for a difference in means between the two groups. In practice, we regressed each variable on a binary indicator equal to one if the student was part of our final sample. For the variables parental tertiary education, number of employed parents, and whether the student is firstborn, p-values were equal to 0.61, 0.21, and 0.26, respectively.

[18] The bivariate OLS coefficient is 0.05, with a p-value of 0.29.

Table 2: Balance Table

Covariates	Control Mean	Difference ($\widehat{\alpha}_{1,k}$)	P-value
Female (=1)	0.366	-0.019	0.790
No Siblings (=1)	0.171	0.015	0.661
Age	16.744	-0.234	0.120
First Born (=1)	0.524	-0.018	0.719
Parent Working (0-2)	1.787	0.002	0.955
Parent University (=1)	0.317	0.032	0.460
University (=1)	0.384	0.068	0.254
STEM University (=1)	0.409	0.012	0.798
STEM Ability (=1)	0.396	-0.054	0.373
STEM Interest (=1)	0.445	-0.032	0.432
GPA	7.133	0.106	0.229
STEM Career (=1)	0.232	0.034	0.494
Fav. Subject STEM (=1)	0.622	-0.107	0.032
Least Fav. Subject STEM (=1)	0.470	-0.049	0.338

Notes: $\widehat{\alpha}_{1,k}$ is a difference in means obtained by regressing each covariate on the binary treatment indicator and on a series of school dummies. The p-value corresponds to its T-test for the regression coefficient $\widehat{\alpha}_{1,k}$ being equal to zero.

6 Empirical strategy

For each outcome variable, we estimate the Intention-to-Treat effect (ITT) with clustered linear regressions. The ITT is the standard estimand in randomized experiments with imperfect compliance (Bloom, 1984; Angrist et al., 1996): it captures the causal effect of being randomly assigned to treatment – i.e., being offered the possibility to access FabLab activities – regardless of whether individual students actually enroll in the courses.[19]

We estimate the following model:

$$Y_{h,i} = \beta_{0,h} + \beta_{1,h} Treat_i + \boldsymbol{\gamma}_h \boldsymbol{X_i} + \varepsilon_{h,i} \tag{1}$$

Where Y_h denotes one outcome in the vector of outcomes $\boldsymbol{Y}$. Our parameter of interest is β_1, estimating the average ITT. Standard errors are clustered at the class level, namely the level of assignment to treatment, following Abadie et al. (2023). $\boldsymbol{X}$ is a vector of baseline covariates, which includes a set of school (stratum) fixed effect dummies, the number of parents who are

[19]Our pre-analysis plan initially foresaw the estimation of a Local Average Treatment Effect by Two-Stages-Least-Squares, Instrumental Variable, but the final number of clusters and characteristics of compliance made that choice unfeasible. A more detailed discussion of deviations from the Pre-Analysis Plan is available in Section C in the Appendix.

active workers, the student's age and three other dummies indicating whether (i) student i is a male; (ii) any of the two parents of student i obtained a university degree; and (iii) student i was the first child. As it is standard to increase statistical power (and to correct for the slight imbalance in liking a STEM subject), we also include baseline outcomes: willingness to attend university, self-reported interest in STEM majors, self-reported ability to pursue a STEM major, and self-reported GPA.

From a policy perspective, the ITT is informative about the effect of offering students the opportunity to participate in FabLab activities. Since in our setting it was only possible to randomize assignment to treatment rather than treatment uptake, in our pre-registration, we planned to estimate the Local Average Treatment Effect (LATE) by Two-Stage Least Squares (2SLS) regression. This estimand would have identified the effect of FabLab courses for compliers, namely those students that, once randomly given the possibility to follow FabLab activities, actually decided to attend them (approximately 38% of the total). As detailed in Section C of the Appendix, our pre-analysis plan also foresaw a substantially larger sample, which would have made 2SLS estimation feasible thanks to a lower variance. Moreover, the observed uptake of FabLab courses makes the ITT estimand more viable and interesting for policy due to likely spillovers within classes. Indeed, 38% of eligible students choose to enroll in FabLab courses, but treated students who do not take up the offer may still be influenced by peers who do. Given that classmates often share similar interests and discuss post-secondary plans, it is plausible that participation by some students affects the outcomes of others. In this context, 2SLS estimates based on individual take-up would mix the direct effect of FabLab participation with peer-driven effects, making it difficult to isolate the impact of attending the course. For these reasons, we combine clustered randomization at the class level with an intent-to-treat estimand, which produce valid causal estimates even in the presence of such spillovers, and can provide relevant insights regarding education policies as they would actually be implemented in the context under study.

We complement the regression-based ITT analyses with individual-level Conditional ITT estimates (CITTs), obtained through honest causal forests (Athey and Imbens, 2016; Athey and Wager, 2019). The method extends random forests to causal inference to predict individual-level causal effects; in our case, CITTs can be interpreted as the intensity of each student's response to the possibility of attending FabLabs, when that possibility is given (predicted also for control units based on the observable covariates). Trees recursively partition the covariate space with a splitting criterion that maximizes heterogeneity in the treatment effect, and the honest property guarantees valid pointwise inference. These estimates serve four purposes. First, despite being exploratory, they corroborate the exploratory heterogeneity and mechanisms' analyses in a non-parametric way, thereby lending these results further credibility and providing additional insights. Second, they allow us to run robustness checks that exclude the presence of biases from the variety of control activities and potential spillovers between treated and control classes. Third, they allow us to compare how compliers (who attend FabLabs) and non-compliers (who do not attend despite the possibility) differ in their potential reaction to FabLab courses, enriching the discussion of the opportunity of making these activities compulsory from a policy perspective. Finally, observing the full distribution of predicted individual causal effects and the importance of each covariate as a driver of effect heterogeneity allows us to better address the limitations arising from having a relatively small sample. Further details on the estimation of causal forests and their results are reported in Section G of the Appendix.

7 Results

7.1 Main analysis

Table 3 summarizes our results on the effect of having the possibility to participate in FabLab courses (ITT estimates). Starting from the outcomes related to university intentions, we find that FabLab courses do not significantly affect the extensive margin—namely, the willingness to pursue higher education (column 1). The estimate is imprecise, hence not significant, due to the limited number of clusters, but it is not small in magnitude: it corresponds to a 3.6 percentage point increase over a baseline of 41% – i.e., a 9% increase. We view this as suggestive evidence that this margin could also be affected in a larger sample, potentially an indication for future studies in this domain. In contrast, we find a stronger and statistically significant effect on the intention to pursue a university degree in STEM (column 3), with an increase of 6.1 percentage points – a 17% rise relative to the baseline. The effect on willingness to enroll in a STEAM faculty (column 2) is close in magnitude, albeit less precisely estimated, resulting in a non-significant estimate.

As discussed in Section 5.2, we included the secondary (and exploratory) STEAM outcomes given the emphasis put on creativity by FabLab courses.[20] We stress that the STEAM outcomes are not the main focus of our contribution: our primary findings concern the STEM outcomes. The STEAM analysis complements these findings by examining whether the creative component of FabLabs may also generate interest in digitally intensive artistic fields – a relevant margin given that our sample includes students from arts-focused schools. The fact that the effect is

[20]By exploratory, we indicate that the STEAM reclassification of fields and occupations was not included in our pre-analysis plan.

stronger for STEM, as opposed to general enrollment, likely reflects the phrasing of the question on field choice: we ask students which area they would choose if they were to go to university. This conditional structure encourages even undecided students to express a preference. As a result, the effect on STEM is larger than the effect on the extensive margin, and it is arguably driven by students who hold a preference for STEM activities despite still being uncertain about whether to enroll in university.

While one might be concerned that conditional intentions do not necessarily translate into enrollment, three considerations are relevant. First, many students in our sample are not in their final year of high school, which likely exacerbates uncertainty about post-secondary plans and may dampen any effect on university enrollment itself. Second, despite this, the magnitude of the effect on the willingness to enroll in a STEM field is substantial: 6.1 percentage points over a baseline of 35% corresponds to a 17% relative increase. Third, existing evidence from randomized experiments that measured both self-reported intentions and actual outcomes supports the view that treatment-induced shifts in stated intentions are informative of actual behavioral changes.[21]

Further support for this interpretation comes from the fact that FabLab students report greater self-efficacy in STEM fields, although their stated interest in STEM subjects does not significantly increase. This pattern maps directly onto the conceptual framework developed in Section 3: FabLab activities, by providing hands-on mastery experiences with digital fabrication tools, shift students' beliefs about their ability to succeed in STEM, while intrinsic preferences shaped by longer-run socialization are less responsive to a single intervention. Accordingly, the increase in conditional intentions to study STEM likely reflects growing aspirations among students who now believe they can succeed in STEM but are still forming their post-secondary plans. We also find evidence of a higher stated probability of pursuing a STEM-related professional career (column 6), with an effect of 7.4 percentage points over a baseline of 18% – i.e., a 40% increase.

Consistent with the third prediction of the framework, when beliefs about one's own ability shift, downstream educational intentions follow. As we discuss in Section 7.2 and Section G of the Appendix, however, the same mechanism does not appear to drive the effect on career aspirations: individual-level CITT correlations reveal that students with larger predicted gains in self-efficacy also experience larger gains in STEM university intentions, but *not* in STEM occupational aspirations. The latter likely respond through a separate channel (plausibly, direct

[21] For example, Breda et al. (2023) collected both survey measures of STEM interest and administrative data on actual enrollment in selective STEM programs, and found that their intervention moved both outcomes consistently. Jensen (2010) measured both perceived returns to education and actual years of completed schooling, finding effects on both margins. Jacob and Wilder (2010) document that educational expectations are strong predictors of actual attainment across multiple US longitudinal datasets.

Table 3: Main Results: Possibility to attend FabLabs, university and occupational attitudes

	University (1)	STEAM Faculty (2)	STEM Faculty (3)	STEM Self-efficacy (4)	STEM Interest (5)	STEAM Occupation (6)	STEM Occupation (7)
Can Attend Fablab	0.036	0.056	0.061^{**}	0.073^{**}	0.031	0.046	0.074^{**}
	(0.026)	(0.033)	(0.030)	(0.033)	(0.035)	(0.032)	(0.033)
Control Mean	0.409	0.451	0.354	0.329	0.360	0.244	0.183

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
All dependent variables are binary indicators. All specifications control for baseline characteristics: the outcomes measured at baseline, school (stratum) fixed effect dummies, the number of parents who are active workers, the student's age and three other dummies indicating whether (i) student i is a male; (ii) any of the two parents of student i obtained a university degree; and (iii) student i was the first child. Standard errors are clustered at the class level and reported in parentheses. STEAM outcomes (columns 2 and 6) are secondary outcomes and were not included in our pre-analysis plan.

exposure to STEM as a creative, applied domain) with different implications for the persistence of the effects, to which we return in Section 8.

Adjusting for multiple hypothesis testing reveals that, for three pre-registered families of outcomes, the significant results on self-reported STEM ability and the willingness to pursue a STEM occupation are robust to a Bonferroni correction at the 10% significance level. The Bonferroni correction is, however, too conservative, as it ignores the positive dependence between our outcomes. Indeed, more positive attitudes towards STEM are instrumental to the pursuit of university studies in STEM and, eventually, STEM occupations. To account for this positive correlation between outcomes both within and across families, we detail and perform the Benjamini and Bogomolov (2014) false discovery rate procedure in Section F of the Appendix. Despite the relatively limited sample size at our disposal, the primary outcomes in each family remain significant at a 10% false discovery rate. As an additional check, in Section D we repeat the main analysis without controlling for baseline covariates, allowing us to verify the absence of major differences between the analysis sample where covariates are observed ($N = 573$) and the full sample where these are partly unobserved ($N = 710$). All results are qualitatively confirmed despite losing precision, as expected, and the coefficient for STEM occupation remains statistically significant at the 5% level.

7.2 Heterogeneity and exploratory mechanisms

The intervention under study consists of granting access to FabLab courses among mandated extra-curricular activities, leaving the take-up of the course up to students to mirror actual conditions faced by a hypothetical policymaker. In this Section, we explore two important dimensions of heterogeneity from a policy perspective: compliance and students' sex. We also investigate their interplay to assess whether FabLabs can mitigate the issue of women self-selecting out of innovative STEM curricula.[22] In the tables that follow, we do not separate STEM and STEAM outcomes for the sake of conciseness, although, as mentioned previously, STEAM outcomes are exploratory.

In the Italian schooling system, high school students are required to earn PCTO credits through internships or extracurricular courses in order to graduate, but the specific activity choice remains generally free, constrained only by the variety of available activities, which is determined autonomously by each school. In our study and in a real-life scenario, FabLab courses become *de facto* compulsory when other alternatives are limited or absent. This is the situation for two schools in our setting (totaling 106 students), where students in treated classes exhibited perfect compliance: i.e., they all attended FabLab courses when given access to them. Imperfect compliance is instead observed for the remainder of the schools (the majority, as expected, totaling 467 students).

Table 4 estimates our main results by OLS separately for schools with imperfect and perfect compliance. Intentions to pursue a STEM profession increase under both imperfect and perfect compliance. In the latter case, the null hypothesis of no effect is only marginally accepted, with a p-value of 0.1, due to the reduced sample size and hence power. Intentions to pursue STEM university studies are also not clearly associated with compliance, with marginally insignificant estimates in both cases. On the other hand, the magnitude of the treatment effect on perceived STEM ability is markedly higher in perfect compliance schools, which seem to drive the overall estimate (0.142 and 0.073, respectively). In Section 2.1, we discussed that students in technical and artistic high school majors are typically those who had lower grades in middle school, and exhibit lower perceived self-efficacy. Possibly then, these results indicate that low self-efficacy at baseline might prevent the take-up of innovative STEM courses, whereas imposing attendance could alleviate self-selection into course enrollment, favoring the exposure to these activities for the students who could benefit the most. These heterogeneity findings and the resulting hypotheses on how compliance impacts mechanisms of action are further supported by causal forest analyses, reported in Section G in the Appendix. In particular, Figure G.5 confirms that access to FabLabs increases self-efficacy among those who displayed lower values at baseline, who need it most. When comparing the distributions of predicted individual causal effects (Figure G.6) between compliers and non-compliers, the only relevant difference concerns self-efficacy in the same direction as OLS heterogeneity results, supporting the notion that non-compliers may achieve the same changes in their attitudes towards STEM studies and occupations if FabLab

[22] While heterogeneity by sex was pre-registered, heterogeneity by compliance is an exploratory analysis, which we included as an alternative to the pre-registered Two-Stages-Least-Squares, Instrumental Variable analysis. A more complete account of deviations from our Pre-Analysis Plan is presented in Section C of the Appendix.

activities were compulsory, and that self-efficacy plays a role in deciding to attend in the first place.

With regard to heterogeneity by sex, Table 5 reveals that the effects documented in the previous Section are largely driven by male students.[23] The comparison between column 1 versus column 8 in Table 5 shows that the effect on the extensive margin—willingness to attend university—is positive for male students (4.7 percentage points), and arguably larger than what we observe in the full sample. For female students, by contrast, the estimate is close to zero and precisely estimated, suggesting that the lack of an average effect is not simply due to limited power.

The same pattern holds when looking at the intention to pursue a university degree in STEM. Male students exposed to the FabLab program are 9.1 percentage points more likely to choose a STEM field if they go to university – a 24% increase relative to their baseline – and the effect is statistically significant at the 10% level. For female students, the estimate is again close to zero. Interestingly, the results for STEAM are more nuanced: we observe no effect on STEM for females, whereas the estimate on STEAM is somewhat larger. While still imprecisely estimated, this suggests that some female students may have developed an interest in FabLab activities primarily through their creative component. Related work documents that FabLab participation also enhances self-reported creativity (Ballerini et al., 2024). Taken together, these results suggest sex-specific pathways through which the program operates: the technical aspects of FabLab activities seem to resonate more with male students, while the creative aspects may be more salient for females.

Similarly, the effects on self-reported ability and interest in STEM are concentrated among male students (columns 11 and 12). For them, perceived ability increases by 8.8 percentage points, nearly identical to the effect estimated in the full sample. Moreover, unlike in the pooled analysis, we also observe an increase in interest in STEM subjects among male students, of similar size. Finally, consistent with what we showed so far, the probability of pursuing a STEM professional career increases only for males (column 14). On the one hand, this is not surprising in light of existing studies that attribute sex differences in STEM careers' take-up to preferences. On the other hand, in the Italian setting, we discussed that male students are less likely to pursue higher education, with especially low numbers of STEM graduates (see Section 2.1).

How do students' sex and compliance interplay? In descriptive terms, Figure A.2 in the Appendix reveals that males are more likely to attend when students are free to choose, lending

[23] Causal forests in Section G in the Appendix replicate this result with a fully data-driven, non-parametric model.

further support to the notion that different preferences for STEM activities drive sex differences. Table 6 reports heterogeneous treatment effects by sex under imperfect compliance.[24] Indeed, intentions to pursue STEM university studies and professions, which were overall statistically significant under imperfect compliance, are driven by male students. This result, in line with the above considerations on heterogeneous effects by compliance, points towards a closer integration of FabLab activities into school planning, in order to foster and encourage participation even from students who would normally opt out of such activities, such as female students.

In terms of mechanisms, a natural follow-up question is whether self-efficacy is the same mechanism through which the intervention affects each outcome. Using the CITT estimates introduced in Section 6, we compare, at the individual level, predicted gains in self-efficacy with predicted gains in STEM-university and STEM-occupation intentions (Figure G.9 in the Appendix). Predicted gains in self-efficacy are positively correlated with predicted gains in STEM university intentions, but *negatively* correlated with predicted gains in STEM occupational aspirations. Self-efficacy therefore appears to be a relevant channel for academic STEM choices, but not for STEM career aspirations, which seem to respond through a distinct mechanism. Possibly, the direct exposure to STEM as a creative, applied domain, which is also compatible with the idea that students may conceive university studies and occupations not as consequential steps, but as potential alternatives. This nuances the conceptual framework in Section 3 and also aligns with the descriptive evidence above that male and female students respond differently to the technical and creative components of FabLab activities, with the latter possibly being more salient for the orientation towards STEM occupations among women.

[24] We are prevented from repeating the same exercise under perfect compliance because of the small sample size and the low representation of female students.

Table 4: Effect of Possibility of Attending FabLab courses on Self-reported University and Career Outcomes: Heterogeneity by Compliance

	Schools with Imperfect Compliance						
	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Can Attend Fablab	0.022	0.066	0.059	0.052	0.033	0.039	0.076*
	(0.032)	(0.040)	(0.036)	(0.041)	(0.039)	(0.041)	(0.040)
Control Mean	0.395	0.460	0.347	0.306	0.306	0.258	0.177
	Schools with Perfect Compliance						
	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Can Attend Fablab	0.076*	0.016	0.058	0.142***	0.018	0.081	0.081
	(0.039)	(0.027)	(0.040)	(0.026)	(0.065)	(0.048)	(0.048)
Control Mean	0.450	0.425	0.375	0.400	0.525	0.200	0.200

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
All dependent variables are binary indicators. Column headers: University = wants to enroll in university; STEAM/STEM Faculty = wants to enroll in a STEAM/STEM faculty; STEAM/STEM Occupation = wants to pursue a STEAM/STEM occupation. All specifications control for baseline characteristics: the outcomes measured at baseline, school (stratum) fixed effect dummies, the number of parents who are active workers, the student's age and three other dummies indicating whether (i) student i is a male; (ii) any of the two parents of student i obtained a university degree; and (iii) student i was the first child. Standard errors are clustered at the class level and reported in parentheses. The number of observations was 467 and 106 for schools with imperfect and perfect compliance.

Table 5: Effect of Possibility of Attending FabLab courses on Self-reported University and Career Outcomes: Heterogeneity by Sex

	Females						
	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Can Attend Fablab	-0.005	0.070	-0.007	0.064	-0.030	0.042	0.047
	(0.057)	(0.051)	(0.044)	(0.049)	(0.029)	(0.034)	(0.030)
Control Mean	0.617	0.417	0.300	0.283	0.350	0.233	0.150
	Males						
	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Can Attend Fablab	0.047	0.028	0.091*	0.088*	0.079	0.070	0.105**
	(0.031)	(0.045)	(0.047)	(0.047)	(0.062)	(0.049)	(0.048)
Control Mean	0.288	0.471	0.385	0.356	0.365	0.250	0.202

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
All dependent variables are binary indicators. Column headers: University = wants to enroll in university; STEAM/STEM Faculty = wants to enroll in a STEAM/STEM faculty; STEAM/STEM Occupation = wants to pursue a STEAM/STEM occupation. All specifications control for baseline characteristics: the outcomes measured at baseline, school (stratum) fixed effect dummies, the number of parents who are active workers, the student's age and three other dummies indicating whether (i) any of the two parents of student i obtained a university degree; and (ii) student i was the first child. Standard errors are clustered at the class level and reported in parentheses.

Table 6: Effect of Possibility of Attending FabLab courses on Self-reported University and Career Outcomes: Heterogeneity by Compliance and Sex of the students

	Schools with Imperfect Compliance – Females						
	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Can Attend Fablab	-0.029	0.078	-0.018	0.066	-0.027	0.029	0.035
	(0.063)	(0.056)	(0.048)	(0.059)	(0.029)	(0.035)	(0.032)
Control Mean	0.596	0.404	0.269	0.231	0.288	0.231	0.135
	Schools with Imperfect Compliance – Males						
	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Can Attend Fablab	0.029	0.035	0.102	0.057	0.096	0.085	0.131**
	(0.041)	(0.059)	(0.063)	(0.059)	(0.072)	(0.066)	(0.064)
Control Mean	0.250	0.500	0.403	0.361	0.319	0.278	0.208

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

All dependent variables are binary indicators. Column headers: University = wants to enroll in university; STEAM/STEM Faculty = wants to enroll in a STEAM/STEM faculty; STEAM/STEM Occupation = wants to pursue a STEAM/STEM occupation. All specifications control for baseline characteristics: the outcomes measured at baseline, school (stratum) fixed effect dummies, the number of parents who are active workers, the student's age and three other dummies indicating whether (i) student i is a male; (ii) any of the two parents of student i obtained a university degree; and (iii) student i was the first child. Standard errors are clustered at the class level and reported in parentheses. The number of observations was 191 and 276 for females and males, respectively.

8 Conclusions

In this paper, we presented the results of a field experiment aimed at fostering interest in STEM majors and STEM careers by intervening in course activities offered to high school students for credit. We conducted an RCT in five Italian schools and randomized 42 classes (with 573 students eventually included in the analysis) to have access to creative STEM activities for credit in their schools, to learn the use of digital fabrication tools. The voluntary uptake of these activities mirrors the functioning of the current Italian schooling system, maximizing the scalability of the intervention. These courses adopted the FabLab non-mnemonic, learn-by-doing pedagogical approach, and emphasized creative thinking to build objects of actual utility. To our knowledge, this is the first study to assess an intervention based on a similar pedagogic approach to improve attitudes towards STEM.

We found a sizeable, statistically significant effect on the intention to pursue a university degree in STEM, with an increase of 6.1 percentage points—a 17% rise relative to the baseline. We also find a higher stated probability of pursuing a STEM-related professional career, with an effect of 7.4 percentage points over a baseline of 18%, and an increase in STEM self-efficacy of 7.3 percentage points (a 22% rise relative to baseline). Our causal forest analysis suggests, however, that these effects do not operate through a single channel: the increase in STEM-university intentions is consistent with growing self-efficacy, as treated students feel more confident in their ability to pursue STEM studies, whereas the increase in STEM career aspirations does not appear to be driven by self-efficacy and likely operates through a distinct mechanism, most plausibly the direct exposure to STEM as a creative, applied domain that broadens students' conception of what STEM-related occupations can look like.

Exploratory heterogeneity analyses also find suggestive evidence of sex-specific pathways through which these activities operate: the technical aspects of FabLab activities seem to resonate more with male students, while females find the creative aspects more salient, although they are less likely to voluntarily enroll, potentially due to lower self-efficacy at baseline. At the same time, we find that increased self-efficacy is driven by schools where all treated students enrolled in the creative STEM courses due to a lack of alternatives.

From a policy perspective, these results suggest that the objective of increasing interest and eventual enrollment in STEM university majors among the least inclined could be achieved by making similar creative STEM activities compulsory within the existing schooling system. Indeed, while several traits of creative learning are mentioned in the latest STEM guidelines for Italian education (Ministero dell'Istruzione e del Merito, 2023), this model has only rarely been adopted in Italian high schools.

An important open question is how persistent these effects will be over time, and the answer may depend on which channel is involved. The shift in self-efficacy that drives the effect on STEM-university intentions is, in our framework, a belief-based change: such changes are revisable as students gather new information and experiences, and may be less durable than preference-based ones unless reinforced. As argued by Yeager and Walton (2011), however, brief interventions that operate through psychological mechanisms such as self-efficacy can produce lasting effects when they feed into recursive processes – for instance, if students who gain confidence in STEM subsequently seek out further STEM experiences, reinforcing the initial shift. The effect on STEM career aspirations, in contrast, plausibly operates through exposure to a previously unfamiliar set of STEM-focused activities, corresponding to a larger perceived set of STEM-related career options. This second channel may prove more durable, as it relates to awareness of available pathways rather than to confidence in one's own ability. Tracking the longer-term persistence of both effects and understanding whether they feed into self-reinforcing processes along the educational trajectory is an important avenue for future research.

Our study faced three main limitations, which could also be addressed by future research in this field. First, the relatively small sample size may limit the statistical power of our estimates and reduce the extent to which our findings can be generalized to broader populations. Moreover, although our causal forest analysis does not detect sizeable changes in effects by school type, the schools and classes participating in this study are not fully representative of the diversity of the Italian high school system, potentially affecting the external validity of the results. Third, the identified mechanisms are grounded in plausible interpretations of the observed patterns but remain largely exploratory. Future research should adopt larger samples and more diverse contexts, building on these early results that reveal the potential of a novel and underexamined type of intervention to yield broader educational gains.

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This study has been pre-registered in AEA RCT registry with the number AEARCTR-0008407 and has been approved by the European University Institute's ethical review board (20210602).

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APPENDIX

A Extra Figures and Tables

Figure A.1: Location of schools included in the study

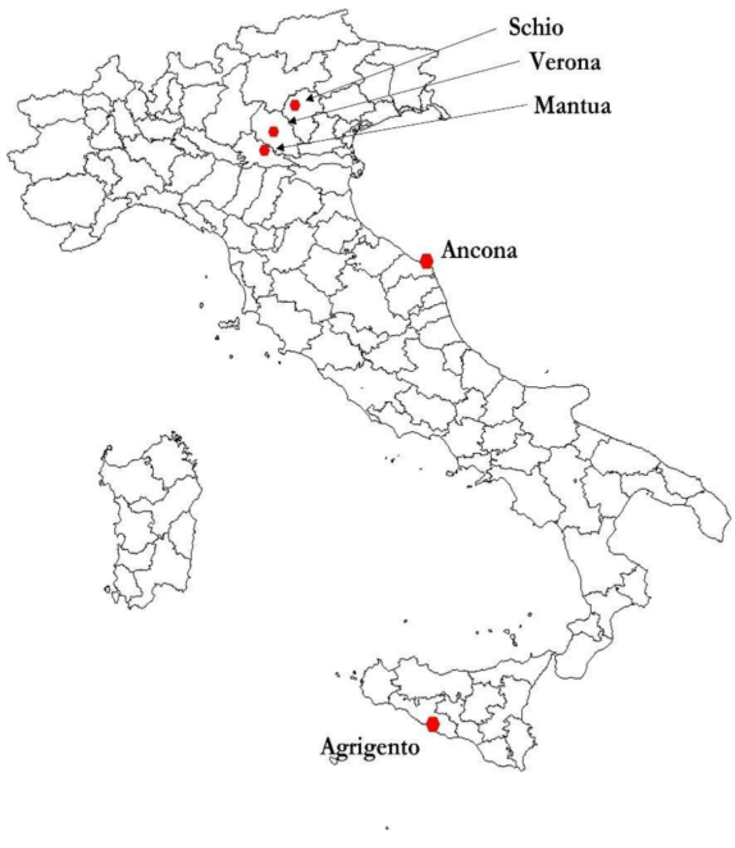


Figure A.2: Compliance by Gender and School

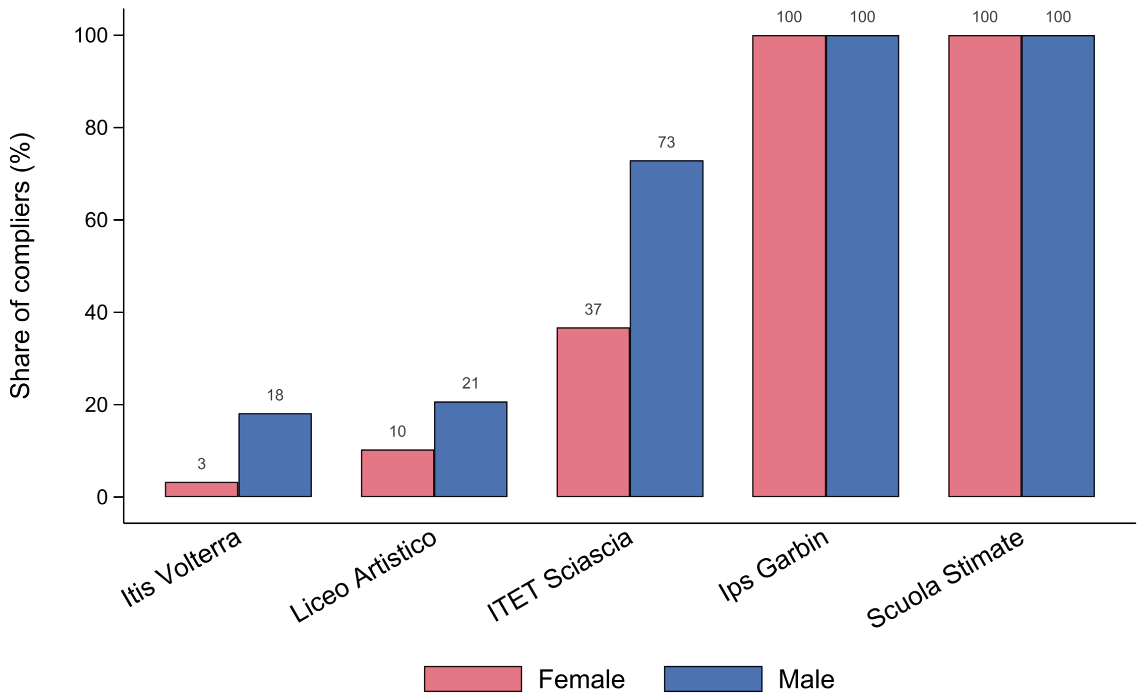


Table A.1: Intended Fields of Study at Baseline by high school major - Males

	Technical major			***Licei***		
	Agrigento	**Schio**	**Ancona**	**Verona**	**Mantua**	**Total**
Architecture	0	0	2	1	4	7
Engineering	4	14	39	9	0	66
Informatics	27	1	17	0	0	45
Science	0	1	12	5	1	19
Art	1	1	4	0	14	20
Agricultural	1	1	1	0	0	3
Economics	6	1	1	6	0	14
Sports Science	4	0	0	0	0	4
Health	3	1	3	0	1	8
Languages	2	0	0	0	0	2
Law	1	0	0	2	1	4
Nursing	4	0	0	0	0	4
Other	3	3	0	1	1	8
Psychology	4	1	1	0	1	7
Don't Know	19	33	77	1	17	147
Total	79	57	157	25	40	358

Notes: Fields of study in bold were classified as STEM. The complete list of STEM courses comprised within these macro categories includes: engineering, information technology, biology, physics, astronomy, mathematics, architecture, earth sciences, computer sciences, and chemistry.

Table A.2: Intended Macro fields of Study at Baseline by school and school major - Females

	Technical major			***Licei***		
	Agrigento	**Schio**	**Ancona**	**Verona**	**Mantua**	**Total**
Architecture	3	0	1	0	13	17
Engineering	1	2	2	3	1	9
Informatics	0	0	1	1	0	2
Science	0	0	3	2	0	5
Art	2	0	1	0	43	46
Agricultural	0	0	0	0	0	0
Economics	8	0	1	1	2	12
Sports Science	1	0	0	0	0	1
Health	1	0	7	12	0	20
Languages	11	0	0	0	2	13
Law	0	0	4	0	0	4
Nursing	2	0	0	0	0	2
Education	7	0	0	0	0	7
Humanity	2	0	0	0	0	2
Other	4	0	0	0	1	5
Psychology	6	0	2	1	6	15
Don't Know	5	0	12	2	36	55
Total	53	2	34	22	104	215

Notes: Fields of study in bold were classified as STEM. The complete list of STEM courses comprised within these macro categories includes: engineering, information technology, biology, physics, astronomy, mathematics, architecture, earth sciences, computer sciences, and chemistry.

B STEM graduates: Italy *vs* Europe

Table B.3: Italy *vs* EU averages: 2023 educational attainments by age and sex of the students

	Panel A: Full working-age population (ages 15-64)					
	% Low educational attainment			**% High educational attainment**		
	Total	**M**	**F**	**Total**	**M**	**F**
EU avg	24.7	26.0	23.4	30.9	28.1	33.7
Italy	36.7	39.3	34.2	19.2	16.1	22.7
	Panel B: Post-university young population (ages 25-34)					
	% Low educational attainment			**% High educational attainment**		
	Total	**M**	**F**	**Total**	**M**	**F**
EU avg	14.5	16.4	12.6	43.1	37.6	48.8
Italy	19.9	23.0	16.6	30.6	24.4	37.1

Notes: The data were obtained by Eurostat (2025a). Low educational attainment corresponds to having completed at most lower-secondary school (levels 0-2 of the International Standard Classification of Education). High educational attainment corresponds to tertiary education, i.e., university studies (levels 5-8 of the International Standard Classification of Education).

C Deviations from the Pre-Analyis plan

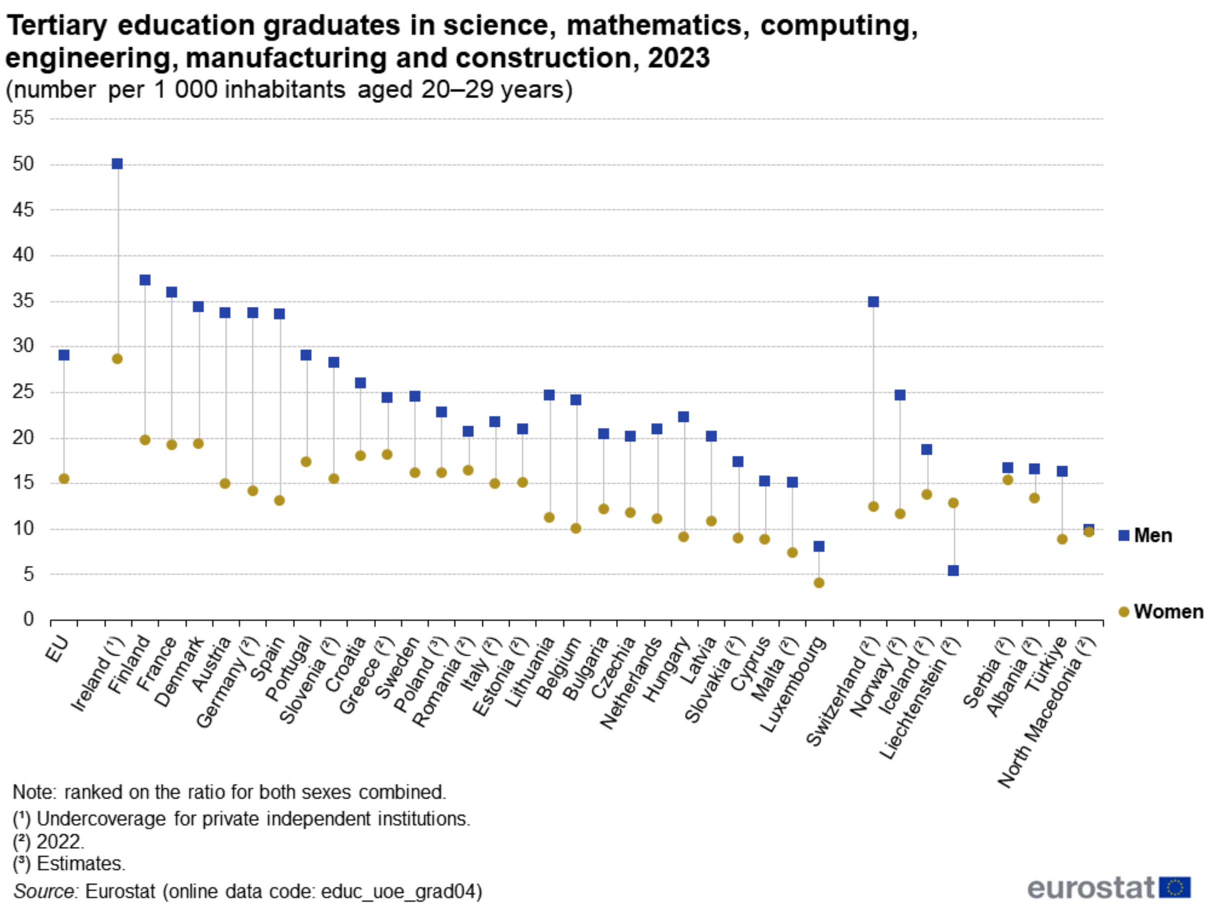
This section details deviations from our pre-registered Pre-Analysis Plan, available on the American Economic Association's registry for randomized controlled trials (as AEARCTR-0011862 Dominici et al., 2025). Where relevant, these deviations have also been reported in the main text, labelling the analyses that were not pre-registered as "exploratory".

A first, necessary clarification is that the Pre-Analysis Plan also covers the outcomes assessed in a companion paper (Ballerini et al., 2024), namely grit and creativity.

The main deviations stem from the fact that we initially planned to involve up to 10 schools (strata) and 60 classes (clusters) in the experiment, for approximately double the sample size – implying also a greater variety of FabLab courses. On top of this reduction from 10 to 5 schools and from 60 to 42 classes, we experienced "inverse" attrition, as explained in the main text, meaning that while all 710 students were present at endline, we could only include 573 in the final analysis because some were absent at the baseline survey, making covariates unobservable.

As a result of sample shrinking and of significant unanticipated variation across schools in

Figure B.3: Shares of young STEM graduates by sex in European countries



Notes: Fields classified as STEM correspond to ISCED-13 codes 05, 06, and 07, corresponding to the classification applied to the sample under study – since code 07 includes architecture.

the degree of compliance (i.e., actually enrolling in FabLab courses when given the possibility), we opted for estimating an Intention-To-Treat effect by OLS rather than the pre-registered Local Average Treatment Effect by 2SLS IV, since the latter would have yielded severely underpowered, imprecise, and thus hard-to-interpret results. To relate to our initial intentions to account for imperfect compliance, we instead included an exploratory heterogeneity analysis where we re-estimate results separately for schools that exhibited perfect and imperfect compliance (Section 7.2). Similarly, these unanticipated sample characteristics prevented us from estimating heterogeneous effects by school, thereby accounting for differences in the specific project of each FabLab course, and limited our possibilities to focus more thoroughly on heterogeneity by sex.

D Removing baseline covariates

These regressions without baseline controls are presented for the sake of completeness but were not pre-registered in our Pre-Analysis Plan.

Table D.4: Possibility to attend FabLabs, university and occupational attitudes: main sample ($N = 573$) and no controls

	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Can Attend Fablab	0.083	0.063	0.065	0.046	0.016	0.057	0.083**
	(0.056)	(0.051)	(0.047)	(0.045)	(0.047)	(0.042)	(0.041)
Control Mean	0.409	0.451	0.354	0.329	0.360	0.244	0.183

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

All dependent variables are binary indicators. All regressions control only for school (stratum) fixed effects, with no baseline covariates. The sample comprises the 573 students for whom we also observe baseline covariates (i.e., present on the day of the baseline survey), i.e. the same sample used in the main analysis. Standard errors are clustered at the class level and reported in parentheses. STEAM outcomes (columns 2 and 6) are secondary outcomes and were not included in our pre-analysis plan.

Table D.5: Possibility to attend FabLabs, university and occupational attitudes: all students ($N = 710$) and no controls

	University	STEAM Faculty	STEM Faculty	STEM Self-efficacy	STEM Interest	STEAM Occupation	STEM Occupation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Can Attend Fablab	0.075	0.052	0.059	0.040	0.032	0.059	0.079**
	(0.051)	(0.048)	(0.044)	(0.043)	(0.040)	(0.040)	(0.039)
Control Mean	0.394	0.428	0.341	0.327	0.337	0.221	0.173

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

All dependent variables are binary indicators. All regressions control only for school (stratum) fixed effects, with no baseline covariates. The sample comprises all students involved in the RCT ($N = 710$), regardless of whether their baseline characteristics were measured with the baseline survey or not. Standard errors are clustered at the class level and reported in parentheses. STEAM outcomes (columns 2 and 6) are secondary outcomes and were not included in our pre-analysis plan.

E Surveys

In this section, we report: (i) the information sheet and consent form received and signed by the parents of all students in the sample; (ii) the English translation of the surveys administered to the students, originally in Italian. Survey section names are reported for the sake of readability, but were not visible to the students. In these documents, we nominate the European University Institute and the University of Florence, which were the authors' affiliations at the time of the RCT implementation.

E.1 Information and consent form for parents

General Information on Data Processing Pursuant to current legislation on the protection of personal data, we inform you that the personal data you provide will be processed lawfully and fairly for purposes related and necessary to the scientific research *"Digital skills, university choice and non-cognitive skills: a Randomized Trial"* of the Fabschool project. The research aims to assess the impact of courses aimed at enhancing digital skills on students' interest in technology, creativity, and professional aspirations.

To achieve this goal, questionnaires or other qualitative research activities will be carried out for scientific research purposes related to satisfaction or self-perception within one's environment, with particular attention to non-formal skills and to any educational/professional opportunities that may have arisen following participation in the courses.

With regard to the scientific research, the data will be processed exclusively by:

- European University Institute (EUI)
- University of Florence

The data controller will be Dr. Martina Francesca Ferracane, who will collaborate with a team of 5 researchers from the European University Institute and the University of Florence.

1. Legal basis — The legal basis for the processing of data is the informed and explicit consent of the data subject, as defined by Articles 5 and 8 of the EUI Data Protection Policy (President's Decision 10/2019), which is based on the principles of the European Data Protection Regulation. In the case of underage participants, consent must be given by those holding parental responsibility. Consent is not mandatory, and the student will not be excluded from the project if they are not interested in participating in the scientific research. Consent may be withdrawn at any time and will remain valid for the entire duration of the activities of the Fabschool project.

The personal data processed are:

- Personal details of the participant and their parents
- Contact details (email/phone) of the participant or their parents (if the participant is underage)
- Data relating to the image and voice of participants obtained through online questionnaires, audio interviews, photos, and videos
- Data relating to the school attended, class and year of enrollment, as well as academic performance

The only sensitive data processed will be the student's gender.

Data will be collected via online questionnaires using the Qualtrics platform and interviews recorded using Microsoft Word. All data are stored on a cloud platform that complies with data protection regulations. Data will be accessible only to researchers directly involved in the scientific research and will not be shared with third parties.

The processing of photos and videos may take place only with the participant's consent, or the consent of a parent (in the case of an underage participant). Consent to photos and videos is voluntary, and refusal will not affect participation in project activities. For a project like Fabschool, an essential aspect is sharing results and best practices through the production of photos and videos to be shared on commonly used channels. If you choose to give consent, you will also contribute to the dissemination of the project among new generations and the local community.

The data collected during the qualitative research and scientific activities of the Fabschool project will be disseminated exclusively in aggregated and anonymous form. This means that the results of an individual participant will never be disclosed in a way that would allow identification. If authorization is requested for the sharing of information that could identify a participant, a separate free and specific consent request will be made.

2. Data retention — The data relating to the scientific research will be kept for a maximum of 3 years after the conclusion and publication of the research, unless different legal or contractual obligations require longer retention. The reason for retaining the data for 3 years is to allow verification of whether the student is interested in participating in further studies to assess the medium- to long-term impact of the project. The data will always be pseudonymized.

3. Rights of the data subject — With regard to your personal data, you have the right to exercise all the rights provided for by Article 16 of the EUI Data Protection Policy. This policy provides that the data subject:

- (a) Is informed if, how, by whom, and for what purpose their data are processed;
- (b) May request the rectification of data if the information is inaccurate or incomplete;
- (c) May request the deletion of data if the processing is unlawful or no longer necessary;
- (d) May block further processing when the conditions under points b) and c) are met.

To exercise your rights, you may at any time contact the Data Controller at the European University Institute at: `martina.ferracane@eui.eu`

4. Requests and complaints — Any requests or complaints may be sent to the European University Institute via email to `martina.ferracane@eui.eu`, copying `data_protection_offer@eui.eu`.

Data Collection and Consent to Processing Having read and understood the above information in all its parts, the adult participant, or the parents in the case of a minor participant:

☐ Authorize the processing of data in the manner indicated in this notice and also authorize requesting from the school the data indicated in point 1.d.

☐ Do not authorize the processing of data in the manner indicated in this notice.

Date: ______________________________

Signature (adult participant or parents): ____

In signing below, in the capacity of having parental responsibility for the minor participant, the undersigned:

- Declares that, at the time of signing, they have parental responsibility for the minor;
- Declares that they sign with the consent of the other parent or have the right to exercise responsibility exclusively, or otherwise have a legally valid title (e.g., a court order) allowing them to act without the other parent's consent;
- Assumes full responsibility for the statements made, releasing the project partners from any liability in the event of false or incomplete declarations or if circumstances arise that are not communicated.

The consent concerns the following participant:

Name: ____

Surname: ____

Date of birth: ____

Home address: ____

Telephone number: ____

E-mail: ____

E.2 Baseline Survey

Hi!

You will have about 30 minutes to complete this questionnaire, though it might take less time. The purpose of the questionnaire is to collect data for a scientific research project of the European University Institute and the University of Florence. We aim to understand whether certain PCTO activities help to develop some personal characteristics.

Your individual data will never be disclosed to anyone (including your school) or published.

If you wish to go back and change an answer, you can do so using the buttons at the bottom of the page.

Studies The next questions are about your school experience and your ambitions for the future.

1. What is your favorite subject? You can indicate up to 2 subjects.
2. Which subjects do you like the least, or find most difficult? You can indicate up to 2 subjects.
    - Italian
    - History
    - Philosophy
    - English or other foreign language
    - Mathematics
    - Chemistry
    - Physics
    - Biology
    - Physical Education
    - Other (specify)
3. How much do you like science subjects, for example computer science, biology, chemistry, and similar subjects?
    - I don't like them at all
    - I tend not to like them
    - I am indifferent
    - I tend to like them
    - I like them
    - I like them very much

4. Do you already know what you would like to do when you grow up? If yes, what? (you can give more than one answer)
   - Yes
   - No
   - I am not sure yet
5. Would you like to go to university?
   - Yes
   - No
   - I am not sure yet
6. If you are considering going to university, which faculty do you think you will enroll in? (you can give more than one answer)
7. If you went to university, would you feel able to follow a STEM course? STEM subjects are those related to engineering, mathematics, statistics, computer science, biology, chemistry, physics, and similar.
   - Yes
   - No
   - I am not sure
8. At university, would you be interested in taking STEM courses? (same definition as above)
   - Yes
   - No
   - I am not sure
9. Why would you be or not be interested in taking STEM courses? (open answer)

Creativity

1. Do you like doing any of the following activities? You can indicate more than one.
   - Composing music
   - Playing an instrument or singing
   - Taking photographs

- Writing and/or making videos
- Painting or drawing
- Writing prose or poetry
- Making sculptures with various materials or ceramic pieces
- Building objects
- Other creative activities (specify)
- None of the above

2. Below you will find several statements that people use to describe themselves. Please indicate the extent to which each statement describes you. There are no right or wrong answers.

    - I think I am a creative person.
    - My creativity is important in defining who I am.
    - I know I can solve even complex problems efficiently.
    - I trust my creative abilities.
    - My imagination and ingenuity set me apart from my friends.
    - I have repeatedly demonstrated that I can face difficult situations.
    - Being a creative person is important to me.
    - I am confident I can tackle problems that require creative thinking.
    - I am good at finding original solutions to problems.
    - Creativity is an important part of who I am.
    - Ingenuity is an important characteristic for me.

    Response scale:

    - Definitely not
    - More no than yes
    - Neither yes nor no
    - More yes than no
    - Definitely yes

3. In the next question, we will ask you to think of as many possible uses as you can for a common object. Example: a brick can be used as a step, a paperweight, a doorstop. If you drill a hole in the middle, it can be used as a flag base. You will have 2 minutes to list all possible uses you can think of.

  - Object: a plastic bottle How could you use a plastic bottle? List all uses you can think of in 2 minutes, separated by commas. (open answer)

4. The next question is similar but presents a situation that requires creative thinking, and you must give a single solution. You have 5 minutes. Imagine that in the future bicycles will no longer need pedals. Think of an original way to reuse a bicycle pedal. The idea should be original in the sense that not many students would think of it. (open answer)

Grit For each statement, indicate how accurately it describes you, compared to most people.

- New ideas and projects often distract me from those I am already working on.
- Obstacles do not discourage me. I do not give up easily.
- I often set a goal but then choose to pursue another.
- I am a hard worker.
- I find it difficult to maintain focus on projects that take more than a few months to complete.
- I finish everything I start.
- My interests change from year to year.
- I am diligent. I never give up.
- I have been obsessed with an idea or project for a short period and then lost interest.
- I have faced and overcome obstacles to win an important challenge.

Response scale:

- Does not describe me at all
- Describes me very little
- Describes me partially
- Describes me fairly well
- This is exactly me

Math How much do you agree with the following statements?

- It is worth making an effort in mathematics because it will help me improve my job prospects.
- Mathematics is important to me because it is necessary for what I want to study later.
- I willingly do mathematics because I enjoy it.
- If I try hard enough, I can get good grades in mathematics.
- I do badly in mathematics regardless of how much I study.
- How well I do in mathematics depends only on me.
- During mathematics lessons, I pay attention.
- I study very hard for mathematics tests.

Response scale:

- Strongly disagree
- Disagree
- Agree
- Strongly agree

Education 2

1. What is the name of your school? (open answer)
2. What is your class/section? (e.g., 3B) (open answer)
3. Approximately, what is your grade point average? (0 to 10 scale)
4. If you have already started PCTO (school-to-work programs), how many hours have you already completed? (number or "I don't know")
5. How many hours of PCTO do you still need to complete? (number or "I don't know")
6. Have you already done PCTO or other extracurricular courses? If yes, briefly describe them.

7. Do you know about FabLabs?
    - Yes
    - No
8. Have you ever attended FabLab courses?
    - Yes, many times
    - Only a few times
    - Never / I don't know them
9. Have you ever participated in CoderDojo activities or similar activities?

Demographics

1. What is your gender identity?
    - Male
    - Female
    - Non-binary
2. What is your biological sex?
    - Male
    - Female
    - Other
3. How old are you? (two-digit number, e.g., 19)
4. Are you an only child?
    - Yes
    - No
5. Are you the firstborn?
6. Does any member of your household run a business, store, or professional office?
7. Which of the following best describes your parents?
    - Both work or are retired

- One parent works or is retired
- Neither parent works/has ever worked

8. To your knowledge, have your parents (one or both) attended university?
   - Yes, both
   - Yes, only my father
   - Yes, only my mother
   - No, neither
   - I am not sure
9. If your father or mother attended university, did they attend any of the following faculties? (multiple answers possible)
   - Engineering
   - Biology
   - Chemistry
   - Physics
   - Computer Science
   - Mathematics
   - Statistics
   - Economics
   - None of the above (specify)
   - I don't know
10. Name and surname (for matching responses with end-of-year survey; will be removed before analysis)

Final Do you have any questions, clarifications, or comments about this questionnaire? (open answer)

E.3 Endline Survey

Hi!

You will have about 30 minutes to complete this questionnaire, though it might take less time. The purpose of the questionnaire is to collect data for a scientific research project of the European University Institute and the University of Florence. We aim to understand whether certain PCTO activities help to develop some personal characteristics.

Your individual data will never be disclosed to anyone (including your school) or published.

If you wish to go back and change an answer, you can do so using the buttons at the bottom of the page.

Creativity

1. How creative do you feel? Scale: 0 = Not at all, 10 = Very much
2. Do you like doing any of the following activities? You can select more than one.
    - Composing music
    - Playing an instrument or singing
    - Taking photographs
    - Writing and/or making videos
    - Painting or drawing
    - Writing prose or poetry
    - Making sculptures with various materials or ceramic pieces
    - Building objects
    - Other creative activities (specify)
    - None of the above
3. Below you will find several statements that people use to describe themselves. Please indicate the extent to which each statement describes you. There are no right or wrong answers.
    - I think I am a creative person.
    - My creativity is important in defining who I am.
    - I know I can solve even complex problems efficiently.
    - I trust my creative abilities.

- My imagination and ingenuity set me apart from my friends.
- I have repeatedly demonstrated that I can face difficult situations.
- Being a creative person is important to me.
- I am confident I can tackle problems that require creative thinking.
- I am good at finding original solutions to problems.
- Creativity is an important part of who I am.
- Ingenuity is an important characteristic for me.

Response scale:

- Definitely not
- More no than yes
- Neither yes nor no
- More yes than no
- Definitely yes

4. In the next question, we will ask you to think of as many possible uses as you can for a common object. Example: a brick can be used as a step, a paperweight, a doorstop. If you drill a hole in the middle, it can be used as a flag base. You will have 2 minutes to list all possible uses you can think of.
   - Object: a can How could you use a can? List all uses you can think of in 2 minutes, separated by commas. (open answer)
5. The next question is similar but presents a situation that requires creative thinking, and you must give a single solution. You have 5 minutes. Imagine the bicycles of the future. How would you improve them? The idea should be original in the sense that not many students would think of it. (open answer)

Grit For each statement, indicate how accurately it describes you, compared to most people.

- New ideas and projects often distract me from those I am already working on.
- Obstacles do not discourage me. I do not give up easily.
- I often set a goal but then choose to pursue another.
- I am a hard worker.

- I find it difficult to maintain focus on projects that take more than a few months to complete.
- I finish everything I start.
- My interests change from year to year.
- I am diligent. I never give up.
- I have been obsessed with an idea or project for a short period and then lost interest.
- I have faced and overcome obstacles to win an important challenge.

Response scale:

- Does not describe me at all
- Describes me very little
- Describes me partially
- Describes me fairly well
- This is exactly me

Studies

1. What is your favorite subject? You can indicate up to 2 subjects.
2. Which subjects do you like the least, or find most difficult? You can indicate up to 2 subjects.
    - Italian
    - History
    - Philosophy
    - English or other foreign language
    - Mathematics
    - Natural and physical sciences (chemistry, physics, biology)
    - Physical Education
    - Arts / Art subjects

- Economics
- Laboratory activities
- Computer Science
- Latin / Greek
- Law
- Mechanics and applied sciences and technologies
- Other (specify)

3. How much do you like scientific and technological subjects, for example computer science, biology, chemistry, technology, and similar?
    - I don't like them at all
    - I tend not to like them
    - I am indifferent
    - I tend to like them
    - I like them
    - I like them very much
4. Do you have ideas about the job you would like to do when you grow up?
5. If yes, what job do you imagine?
6. Even if you do not yet know what job you will do, can you share some ideas?
7. Imagine yourself as an adult. Realistically, what job do you think you will do?
8. If the job you would like to do is different from the one you realistically think you will do, why?
    - I don't think I would have the right skills
    - I don't think I am suited for it
    - It's hard to find this kind of job in Italy
    - It requires training that is too expensive
    - The job I would like is the same as the one I realistically think I will do
    - Other (specify)
9. Would you like to go to university?

- Yes
- No
- I am not sure yet

10. If you are considering going to university, which faculty do you think you will enroll in? (you can give more than one answer)
    - Architecture
    - Fine Arts Academy
    - Economics
    - Law
    - Engineering
    - Computer Science and Technologies
    - Nursing and health professions
    - Classical and Modern Literature
    - Languages
    - Medicine
    - Psychology
    - Education Sciences
    - Sports Sciences
    - Natural, physical and mathematical sciences
    - Communication Sciences
    - Political Science
    - ITS (post-secondary technical specialization schools, no degree awarded)
    - Other
11. At university, would you be interested in taking STEM courses? STEM subjects are those related to engineering, mathematics, statistics, computer science, biology, chemistry, physics, and similar.
    - Yes
    - No
    - I am not sure

12. Why would you be or not be interested in taking STEM courses? (open answer)

13. If you went to university, would you feel able to follow a STEM course? (same definition as above)

    - Yes
    - No
    - I am not sure

Math How much do you agree with the following statements?

- It is worth making an effort in mathematics because it will help me improve my job prospects.
- Mathematics is important to me because it is necessary for what I want to study later.
- I willingly do mathematics because I enjoy it.
- If I try hard enough, I can get good grades in mathematics.
- I do badly in mathematics regardless of how much I study.
- How well I do in mathematics depends only on me.
- During mathematics lessons, I pay attention.
- I study very hard for mathematics tests.

Response scale:

- Strongly disagree
- Disagree
- Agree
- Strongly agree

Education 2

1. In which city is your school located? (list of options given in original)
2. What is the name of your school? (list of options given in original)
3. What is your class/section? (list of options given in original)
4. Approximately, what is your grade point average? (0 to 10 scale)
5. Have you already done PCTO (school-to-work programs) or other extracurricular courses? If yes, briefly describe them.
6. Do you know about FabLabs?
    - Yes
    - No
7. Have you ever attended FabLab courses?
    - Yes, many times
    - Only a few times
    - Never / I don't know them
8. Have you ever participated in CoderDojo activities or similar activities?

Demographics

1. What is your gender identity?
    - Male
    - Female
    - Non-binary
2. What is your biological sex?
    - Male
    - Female
    - Other
3. How old are you? (two-digit number, e.g., 19)

4. Are you an only child?
    - Yes
    - No
5. Are you the firstborn?
6. Does any member of your household run a business, store, or professional office?
7. Which of the following best describes your parents?
    - Both work or are retired
    - One parent works or is retired
    - Neither parent works/has ever worked
8. To your knowledge, have your parents (one or both) attended university?
    - Yes, both
    - Yes, only my father
    - Yes, only my mother
    - No, neither
    - I am not sure
9. If your father or mother attended university, did they attend any of the following faculties? (multiple answers possible)
    - Engineering
    - Biology
    - Chemistry
    - Physics
    - Computer Science
    - Mathematics
    - Statistics
    - Economics
    - None of the above (specify)
    - I don't know
10. Name and surname (for matching responses with baseline survey; will be removed before analysis)

Final Do you have any questions, clarifications, or comments about this questionnaire? (open answer)

F Multiple Hypothesis Testing

As detailed in Section 5.2 of the main text, we classified our primary outcomes into three families: (i) University choice (two measures: willingness to attend university and willingness to enroll in a STEM faculty), (ii) Attitudes (two measures: self-reported interest and ability in STEM), and (iii) Occupation (one measure: willingness to pursue a STEM occupation).

These outcomes are all positively correlated both across and within families: a Bonferroni correction is therefore too conservative, as it assumes independence between outcomes. Nevertheless, of our three significant estimates, both the result for self-reported STEM ability (T stat = 2.21) and the result for the willingness to pursue a STEM occupation (T stat = 2.24) are robust to a Bonferroni correction for three families of outcomes at the 10% significance level (critical value 2.12).

However, to account for multiple hypothesis testing while acknowledging the positive dependence of outcomes, we implemented a hierarchical false discovery rate (FDR) control following the Benjamini and Bogomolov (2014) procedure. This is a less conservative two-step approach, relative to Bonferroni, that controls the expected proportion of false discoveries while accounting for the correlation structure both across and within outcome families. As a first step, for each family, we first computed a *class-level* p-value using the Simes (1986) method, which orders the m p-values within a family as $p_{(1)} \leq \cdots \leq p_{(m)}$ and defines the Simes p-value as

$$p_{\text{Simes}} = \min_{1 \leq k \leq m} \left\{ \frac{m}{k}\, p_{(k)} \right\}.$$

This procedure controls the family-wise error rate under independence and certain forms of positive dependence, making it appropriate for positively correlated outcomes within a family.

We then applied the Benjamini–Hochberg (BH) procedure to the three class-level p-values to control the FDR across families. The BH procedure consists of: (1) sorting the M p-values in ascending order, $p_{(1)} \leq \cdots \leq p_{(M)}$; (2) finding the largest rank k such that

$$p_{(k)} \leq \frac{k}{M}\, q,$$

where q is the desired FDR level; and (3) rejecting the null hypotheses for all p-values with rank $i \leq k$. At $q = 0.10$ and $M = 3$, the BH thresholds are 0.0333, 0.0667, and 0.10 for ranks

1, 2, and 3, respectively. Since all three Simes p-values (0.025, 0.054, 0.084) are below their respective thresholds, all three families were selected for within-family testing.

For the subset of families selected at the class level (here, all three), we applied the BH procedure again within each family to the original outcome-level p-values. Following Benjamini and Bogomolov (2014), the within-family BH was applied at a level $(R/M) \cdot q$, where R is the number of families selected in the first step and M is the total number of families. With $R = 3$ and $M = 3$, this yields the same target level of $q = 0.10$ within each family. This second stage identified one significant outcome in each family: *Wants to enroll in STEM faculty* (*University choice*), *Self-reported STEM ability* (*Attitudes*), and *Wants to pursue STEM occupation* (*Occupation*).

Table F.6: Hierarchical FDR adjustment by outcome family

Family	Raw p-values	Family Simes' p-value	Significant outcome at FDR 10%
University choice	0.166, 0.042	0.084	Wants to enroll in STEM faculty
Attitudes	0.375, 0.027	0.054	Self-reported STEM ability
Occupation	0.025	0.025	Wants to pursue STEM occupation

Notes: The raw p-values refer to our main results displayed in Table 3, in the main text, where no multiple hypothesis correction was applied. Simes' p-values are computed within each family. "Significant" indicates rejection of the null (i.e., coefficient equal to zero) after class-level and within-family Benjamini–Hochberg adjustments, controlling the false discovery rate (FDR) at 10%.

G Estimation of Conditional ITTs via causal forests

This section describes the implementation of the causal-forest analysis introduced in Section 6, and reports the descriptive output that supports the exploratory mechanisms discussion and the robustness checks in Section 7.2. Intuitively, a causal forest can be thought of as a flexible, data-driven version of subgroup analysis. Rather than asking the researcher to specify in advance which covariate splits to test (e.g., by sex, by school type), the algorithm itself splits the covariate space and identifies the dimensions along which responses to the treatment (access to FabLabs) differ most. The output is, for every student in the sample (including students in control classes) a predicted individual-specific treatment effect: a CITT estimate $\widehat{\tau}(\boldsymbol{X_i})$ that captures the intensity of the predicted response to FabLab access for a student with baseline characteristics $\boldsymbol{X_i}$. In addition, causal forest estimation provides variable importance, a statistic that indicates the relative importance of each covariate in $\boldsymbol{X}$ in determining treatment effect heterogeneity.

More formally, the algorithm builds a large ensemble of regression trees (Athey et al., 2019), each grown on a random subsample of the data. Within every tree, splits are chosen to maximize heterogeneity in the estimated treatment effect across the resulting child nodes; in other words, the forest *looks for* regions of the covariate space where the treatment matters more, and regions where it matters less. The CITT estimate for student i is then obtained by aggregating, across the trees in the forest, the within-leaf treatment-effect estimates from the leaves into which i's covariate vector falls. The *honest* property (Wager and Athey, 2018) is what allows pointwise inference: within each tree, the data are split into two halves, one used to choose where to make the splits, and one used to estimate the treatment effects within the resulting leaves. This decoupling prevents the splits from being optimized in ways that would overfit the within-leaf estimates, and delivers valid asymptotic standard errors for each individual CITT estimate $\widehat{\tau}(\boldsymbol{X_i})$.

We estimated one causal forest for each primary outcome of interest, using the `grf` R package: STEM self-efficacy, intention to enroll in a STEM university course, and intention to pursue a STEM occupation. Each forest is grown with 10,000 trees, the default honesty fraction of 0.5, and the default minimum leaf size of 5. The covariates fed to the forest are the same as those used in our main regression specification, augmented with the student's grade in school and an indicator for whether the school's main track is technical as opposed to *licei*. All forests use the same random seed for reproducibility.

Drivers of heterogeneity. Variable importance shows which baseline characteristics drive the heterogeneity in predicted treatment effects, and is measured by how often the covariate is selected as a splitting variable across the trees of the forest, relative to others. In our case, it is particularly useful to discern which baseline characteristics matter more in a setting where many covariates are correlated (e.g., the type of school track and the share of females), making it more difficult to identify drivers clearly from OLS regressions with interactions. While the correlation affects causal forests too, it does so to a lesser degree: the splitting algorithm has room to discern, by looking at individual observations, where covariates that are correlated on average may still assume different values.

Figure G.4 reports, for each outcome, the relative importance of each baseline covariate. As a complementary view, Figure G.5 reports the coefficient of a univariate regression of the predicted CITT on each standardized baseline covariate; this provides a compact summary of the *direction* in which each covariate moves the predicted response. Across all three outcomes, the student's GPA, age, and the outcomes measured at baseline emerge as the most influential characteristics, with patterns broadly consistent with the conceptual framework developed in Section 3. In particular, access to FabLabs appears to improve self-efficacy among those who had lower levels, or in other words, where most needed. The correlation changes sign when looking at the intention to pursue a STEM occupation, supporting the notion that increased self-efficacy may drive interest in pursuing a STEM university faculty but not necessarily a STEM occupation. Interestingly, gender only appears to be a relevant driver of the effect on the intention to pursue a STEM occupation. Age, which is correlated but not entirely overlapping with grade, is always an important driver of heterogeneity. However, the direction of the association varies with the outcome: the perceived ability to pursue STEM university studies (self-efficacy) improves more among older students in higher grades, whereas the opposite happens for the intention to pursue a STEM occupation, which mostly increases among younger students in lower grades. Possibly, this difference can be rationalized by considering that intentions are set, in abstract, already in the first grades, whereas self-efficacy takes longer to adjust, as the thought of actually pursuing a specific path becomes more concrete closer to graduation. Another interesting result is that coming from a technical school rather than a *liceo* is never an important predictor of treatment heterogeneity: if anything, the direction of the correlation is against the expectations, since students in technical tracks (not university-oriented) experience more positive effects on self-efficacy and their intentions to pursue STEM university courses.

Compliers vs non-compliers. Section 7.2 in the main text presented heterogeneity by compliance regime, comparing schools where alternatives to FabLabs were available to those where attendance was *de facto* compulsory. The CITT estimates allow a complementary comparison at the individual level: among students randomly assigned to treatment, do those who chose to attend FabLab courses (*compliers*) and those who did not (*non-compliers*) differ in their predicted response to FabLab access?

Figure G.6**A.** plots the distribution of CITTs among assigned students separately for compliers and non-compliers, by outcome. For the intention to enroll in a STEM university course and the intention to pursue a STEM occupation, the two distributions overlap substantially, with non-compliers exhibiting higher variance hence lower predictability: the predicted response to FabLab access, however, is essentially the same regardless of whether students chose to attend or not. This is substantively relevant from a policy standpoint. It implies that, on these two outcomes, non-compliers would experience treatment effects of magnitude comparable to compliers if attendance were made compulsory, and the average ITT on these two outcomes would not change much under mandatory attendance.

The picture is different for self-efficacy: the distribution of predicted CITTs is shifted downwards for non-compliers relative to compliers, indicating that those who chose not to attend would have smaller predicted gains in self-efficacy if exposed. Combined with the OLS evidence in Section 7.2, where perfect-compliance schools display the largest treatment effect on self-efficacy (0.142 *vs.* 0.073 in imperfect-compliance schools, see Table 4), this pattern is consistent with a dual role of self-efficacy. On the one hand, low baseline self-efficacy plausibly discourages students from voluntarily enrolling in a novel STEM activity, but those same students would have the most to gain from being exposed (as seen in Figure G.5). On the other hand, deciding not to attend is associated with lesser potential improvements in self-efficacy, further amplifying the gap. Mandating attendance would therefore close this gap by reaching students who would otherwise self-select out, consistent with the larger self-efficacy effects observed in perfect-compliance schools.

Figure G.6**B.** reports the same comparison with GPA (the strongest driver of heterogeneity according to Figure G.4) on the horizontal axis. Compliers and non-compliers occupy broadly similar regions of the GPA–CITT space: within each outcome, the structural relationship between baseline GPA and the size of the predicted treatment effect does not depend on whether students self-selected into attendance. This further supports the view that the differential between compliers and non-compliers in self-efficacy is not the result of a structural association between GPA and compliance, but mostly reflects the policy-relevant differences between schools where compliance was *de facto* compulsory and those where it was fully voluntary.

Robustness: variation in alternative control activities. By producing the entire distribution of predicted individual causal effects, causal forests also allow us to compare the intensity of the effects across different control activities offered by different schools. It should be noted, however, that this is a robustness check rather than another heterogeneity analysis: in fact, since control activities mostly vary at the school and grade level, the variety in control activities is already absorbed in our main OLS specification by controlling for school and grade fixed effects. Figure G.7 compares the distribution of estimated CITTs across the three types of alternative activities offered to control classes (nothing, internship, and safety/finance courses). The CITT distributions overlap substantially across the three groups, providing direct evidence that the type of alternative offered to control students does not systematically shift the predicted FabLab access effect.

Robustness: between-arm spillovers. Although cluster randomization assigns entire classes to treatment or control and thereby rules out cross-class contamination, in principle, students in control classes could be indirectly affected by treated peers from other classes in the same school. CITT estimates help uncover this type of spillover because, if present, one would expect control students to be affected more in schools where a higher share of students are treated. Even though the proportion of classes assigned to treatment is fixed, the share of students can vary because class sizes differ. As a descriptive check, Figure G.8 plots: in Panel (A), a scatterplot with a regression line detailing the correlation between control-individuals CITTs and the share of treated students in their school; in Panel (B), the distribution of estimated CITTs among control-class students, grouped by terciles of the share of their school's students assigned to treatment. The CITT distribution does not change significantly with the school-level treatment intensity, consistent with the absence of meaningful between-arm spillovers in our setting.

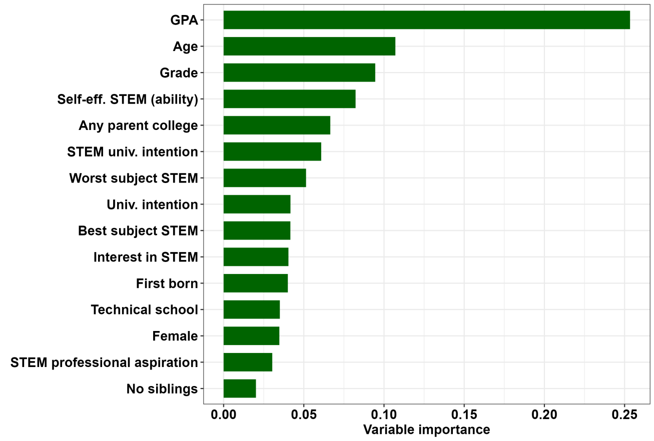
Self-efficacy as a mechanism: cross-outcome correlations. The conceptual framework in Section 3 predicts that FabLab activities operate primarily through self-efficacy: mastery experiences shift students' beliefs in their own ability to engage with STEM, which then feeds into both the intention to enroll in STEM university courses and the intention to pursue STEM careers. Under this prediction, students with larger predicted gains in self-efficacy should also display larger predicted gains in the two intention outcomes. The CITT estimates allow a direct test of this prediction at the individual level: for each student we have a predicted treatment effect on each outcome, and we can examine the cross-outcome correlations.

Figure G.9, Panel (A), plots the predicted CITT on STEM university intentions against the predicted CITT on self-efficacy, with an OLS regression line and its slope and standard error annotated on the plot. The association is positive but small in magnitude: students predicted to gain more in self-efficacy are also predicted to gain marginally more in STEM university intentions, consistent with self-efficacy operating as a partial channel through which FabLab activities shift academic STEM choices.

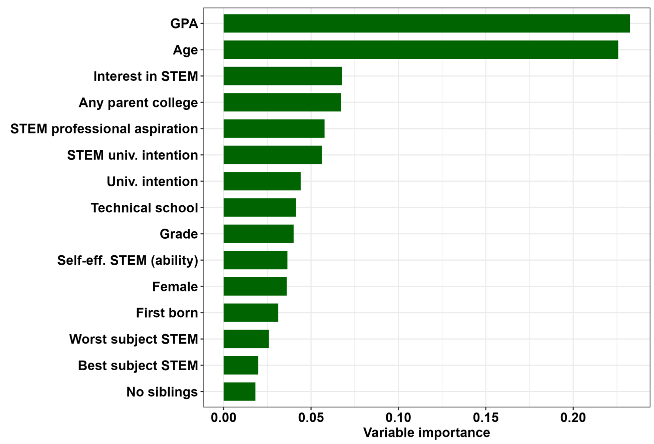
Panel (B) repeats the exercise with STEM occupational aspirations on the vertical axis. The association is now *negative* and of moderate magnitude: students predicted to gain more in self-efficacy are not predicted to gain more in occupational aspirations, and if anything the relationship runs in the opposite direction. This pattern indicates that the channel through which FabLab activities shift career intentions is distinct from the self-efficacy channel that drives the university-intention result. A plausible interpretation is that occupational aspirations are shaped by direct exposure to STEM as a creative, applied domain, broadening students' conception of what STEM activities and careers can look like rather than by an upward revision in beliefs about their own ability. We discuss the implications of this distinction for the conceptual framework in the mechanisms paragraph at the end of Section 7.2.

Figure G.4: Variable importance in CITT estimation

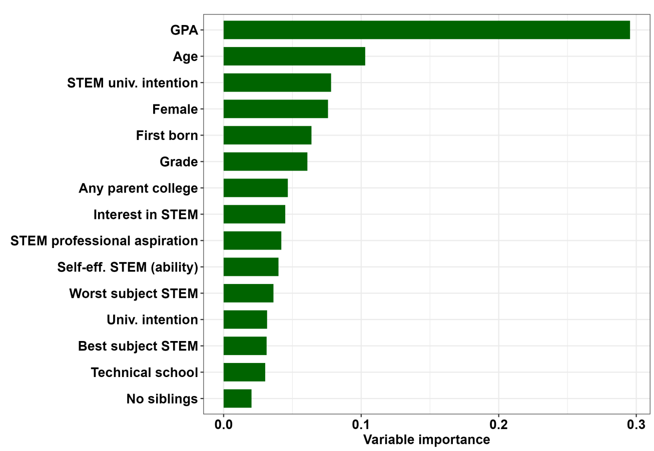
(A.) Outcome: self-efficacy



(B.) Outcome: Intention to pursue a STEM occupation



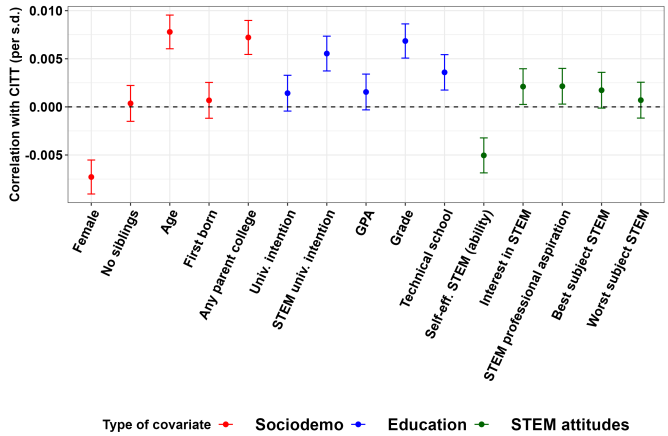
(C.) Outcome: Intention to pursue STEM university studies



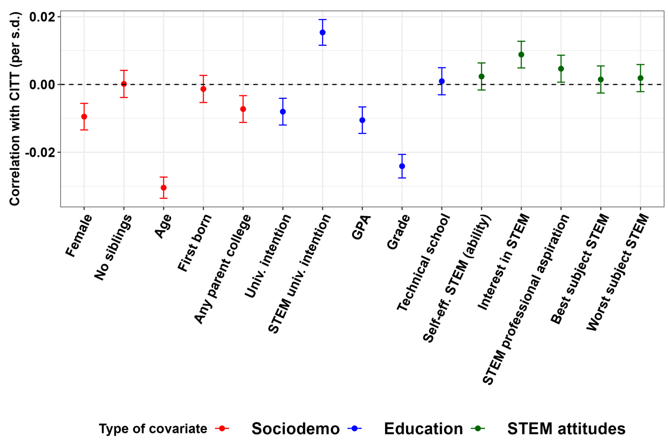
Notes: The three panels show the estimated variable importance of each covariate in determining heterogeneity in CITT estimates (splits in causal forest estimation). Absolute values of variable importance have no direct interpretation, but only relative to other covariates.

Figure G.5: Association between CITT intensity and baseline characteristics

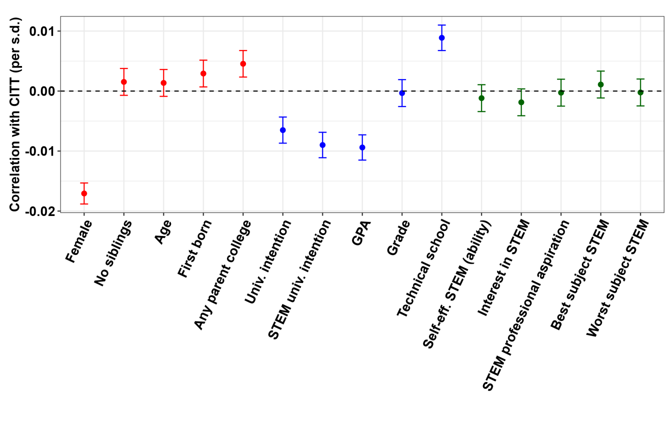
(A.) Outcome: self-efficacy



(B.) Outcome: Intention to pursue a STEM occupation



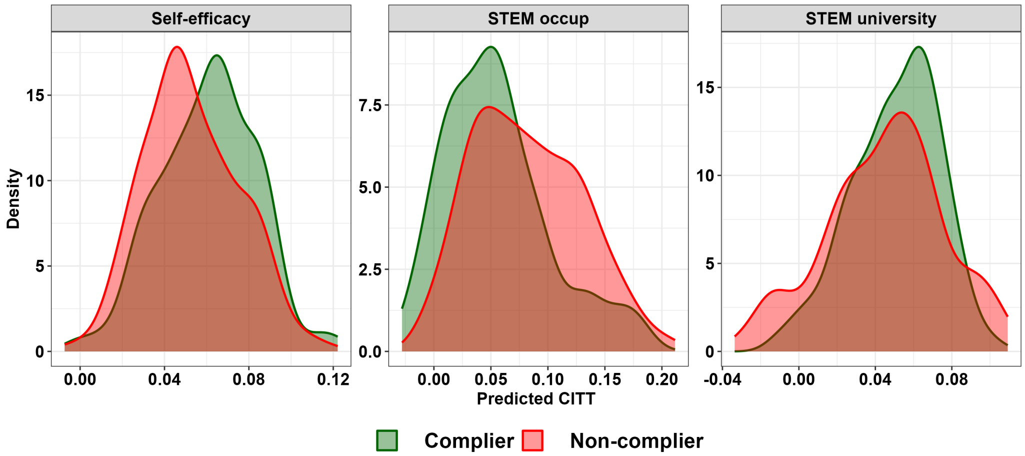
(C.) Outcome: Intention to pursue STEM university studies



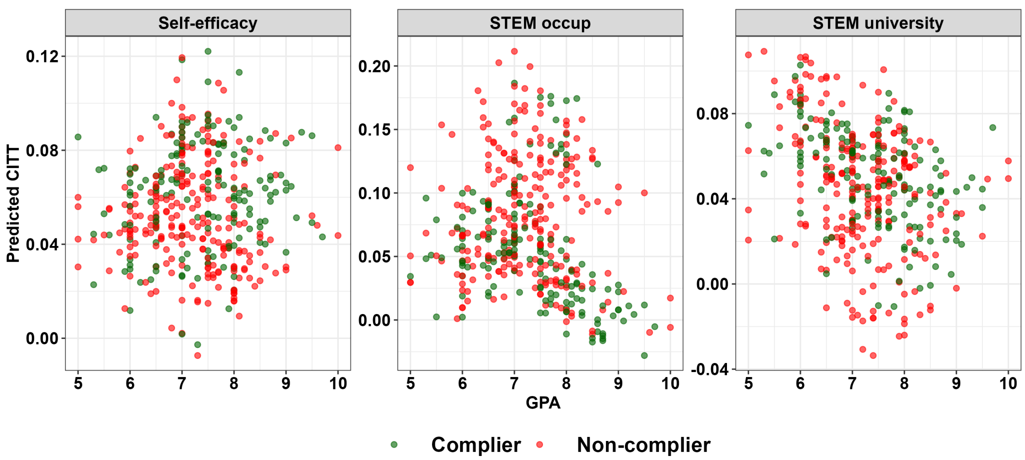
Notes: the figures show, for each outcome, the correlation between individual CITT estimates from causal forests and standardized baseline covariates, estimated by univariate linear regressions. Magnitudes should be interpreted as covariates' standard deviations. Bars indicate 95% confidence intervals.

Figure G.6: Predicted CITT by compliance

(A.) CITT distributions by compliance status

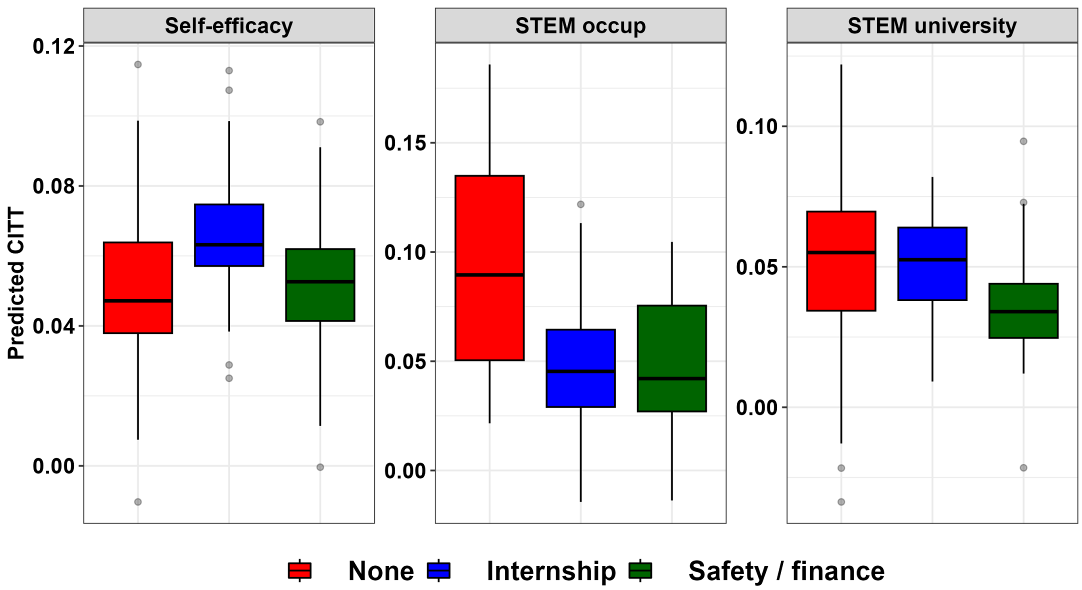


(B.) CITT by compliance and best predictor (GPA)



Notes: Panel (A) shows the distribution of predicted Conditional Intention-to-Treat (CITTs) by compliance status among treated subjects offered FabLab courses, separately by outcome. Panel (B) shows scatterplots correlating individual CITTs with the main predictor of CITT magnitude (i.e., causal effect heterogeneity), namely GPA at baseline.

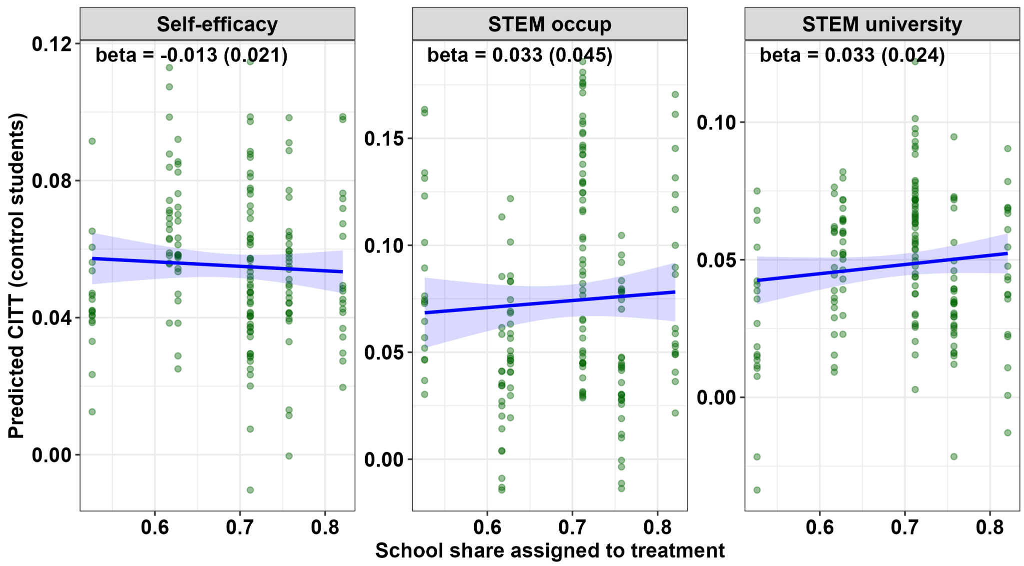
Figure G.7: CITTs by control activity



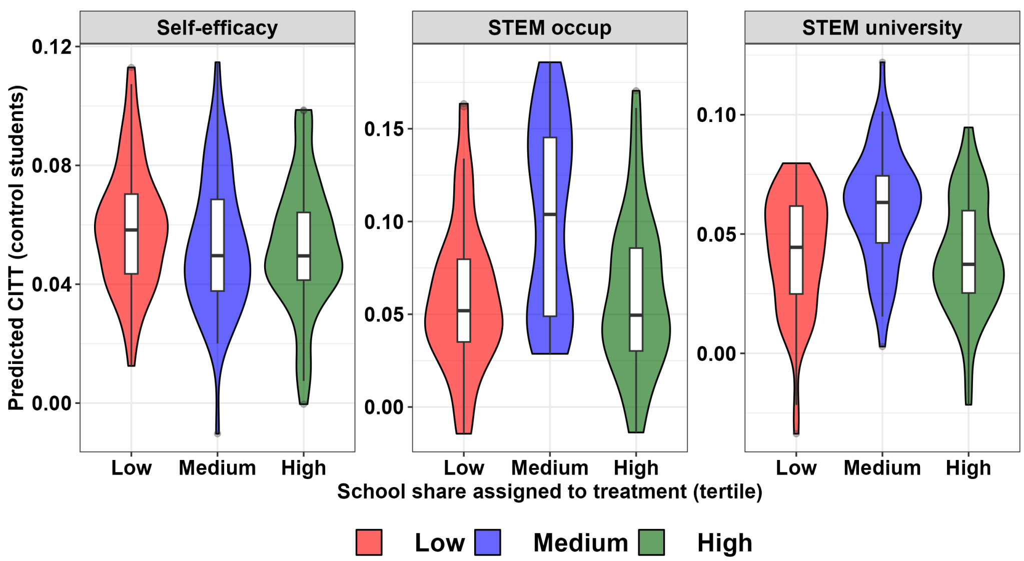
Notes: the figure shows, for each outcome, the distribution of predicted individual causal effects (CITTs) by the type of activity offered to control classes. These can be either absent ("None"), a non-STEM internship ("Internship"), or a course of safety on the job and basic financial literacy ("Safety/finance"). Distributions are displayed with boxplots displaying the IQR and median.

Figure G.8: Informal spillover test: spillover by students' treated share

(A.) Scatterplots and regression test



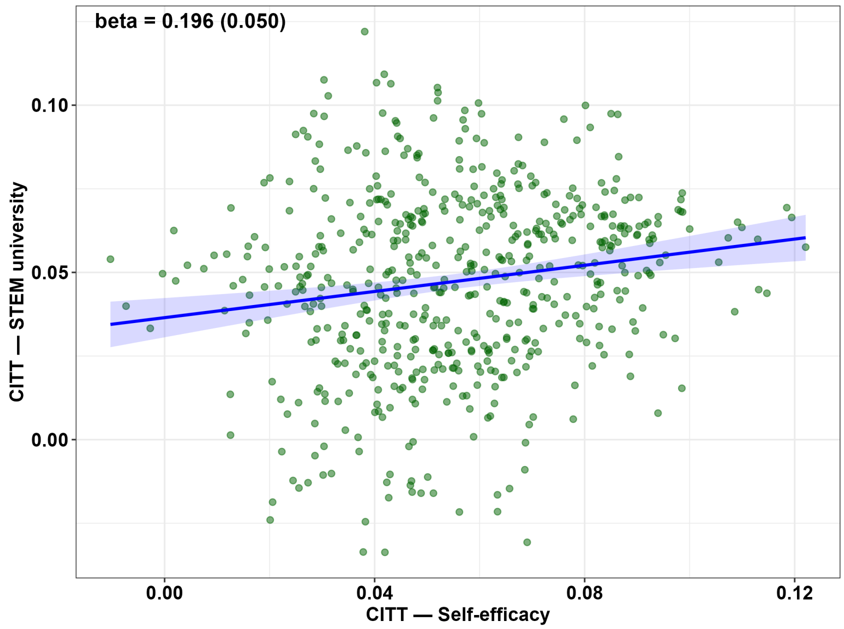
(B.) CITT by share of treated students' tertiles



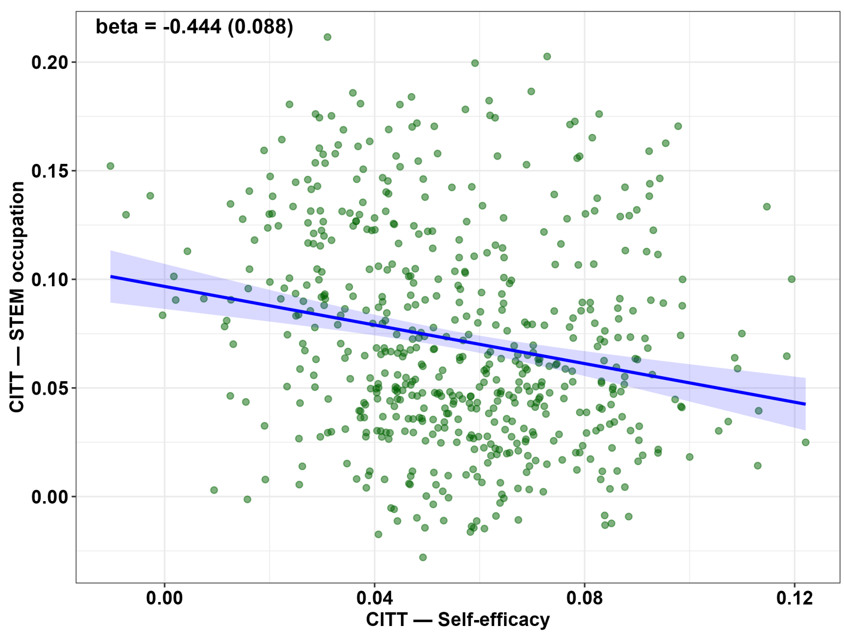
Notes: Panel (A) shows, for each outcome and among control students only, the correlation between predicted CITT and the share of treated students in the same school, estimated by OLS. Since assignment to treatment is at the class level and classes have different sizes, this creates variation in the share of treated students: in the presence of spillovers from treated to control students, one should expect a higher CITT for control students when the share of treated students is higher. Panel (B) summarizes the distribution of CITTs, separating between schools with a share of treated students in the first, second, and third terciles.

Figure G.9: Correlations between treatment effects' intensities on different outcomes

(A.) Self-efficacy and STEM university



(B.) Self-efficacy and STEM occupation



Notes: the two Panels show scatterplots with a linear regression line detailing the correlation between: (A) CITTs on self-efficacy and on intentions to pursue STEM university courses; (B) CITTs on self-efficacy and on intentions to pursue a STEM occupation.